



Master's in Specialized Economic Analysis

“Institutional Quality and the Fiscal Transmission of Energy Shocks: Evidence from Oil-Importing Economies ”

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Abstract

This paper examines whether institutional quality conditions fiscal responses to exogenous oil supply shocks in oil-importing economies. Using state-dependent local projections with a composite institutional index on 29 countries (1974–2024), I find that primary balance responses diverge sharply across institutional regimes. A fiscal decomposition reveals that the asymmetry operates through revenue and debt channels rather than expenditure adjustment. A smooth-transition extension shows that property rights, rule of law, and bureaucratic quality act as near-discrete thresholds for international capital market access, underscoring the centrality of state capacity for fiscal space. Institutional credibility thus emerges as the primary mechanism through which institutional quality shapes fiscal resilience to energy shocks.

Keywords: Fiscal policy, political economy, energy commodities

Resumen

Este trabajo analiza si la calidad institucional condiciona la respuesta fiscal ante shocks exógenos de oferta petrolera. Mediante state-dependent local projections y un índice compuesto de calidad institucional para 29 países entre 1974 y 2024, se muestra que la respuesta del saldo primario varía significativamente según el contexto institucional. La descomposición fiscal indica que estas diferencias se canalizan principalmente a través de los ingresos públicos y la dinámica de la deuda. Asimismo, una extensión smooth transition identifica umbrales vinculados a los derechos de propiedad, el Estado de derecho y la calidad burocrática, subrayando la importancia de la capacidad estatal y la credibilidad institucional para la resiliencia fiscal.

Palabras clave: Política fiscal, economía política, commodities energéticos

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1 Introduction

“The history of economic progress can be written in terms of the advance in the control of energy” (Cottrell 1955). By this logic, the history of economic instability can be written in terms of the failure to control energy, or to respond to its shocks. Oil, the most globally traded and politically charged energy commodity, has repeatedly demonstrated that countries do not merely consume it; their economic and political trajectories are often shaped by it. For oil-importing economies, a sudden disruption in oil supply is not just a price shock, it is a test of the fiscal and institutional architecture that governs public spending, taxation, and sovereign borrowing.

A large body of political-economy theory argues that institutional quality, underpinned by state capacity, accountability, and the enforcement of the rule of law, shapes each of these margins. Strong institutions may therefore lie at the crux of a country’s ability to respond to oil supply shocks, conditioning fiscal responses in oil-importing nations.

This study exploits a measure exogenous oil-supply news shocks developed by Känzig (2023), and embeds it in a state-dependent local projection framework to test the central hypothesis: does institutional quality condition a country’s fiscal response to oil supply shocks? Specifically, does the evidence support the three main political-economy mechanisms, the voracity effect, the state-capacity hypothesis, and the fiscal space hypothesis? The institutional effect is predicted to operate primarily through the debt and revenue channels, while the voracity-driven expenditure response may be attenuated in the context of an adverse shock for oil importers.

The analysis proceeds in four stages: (i) a binary-threshold local projection that splits the sample into low- and high-institution regimes; (ii) a smooth-transition logistic function that allows the fiscal response to vary continuously with institutions; (iii) a decomposition of the primary balance into expenditure, revenue, and debt to test each theoretical channel; and (iv) a disassembly of the composite institutional index into its six sub-indices to identify which dimensions matter most for each fiscal margin.

The empirical analysis finds that institutions do indeed shape fiscal responses to oil-supply shocks, and it reveals a clear hierarchy among the channels. The debt margin displays the largest and most consistent institutional gradient, followed by revenue. Strikingly, property rights, rule of law, and bureaucratic quality operate as thresholds, dividing countries into those that can borrow and those that cannot, a pattern that directly supports the debt-intolerance and fiscal-space hypotheses. These results offer a nuanced, evidence-based account of how institutional quality determines the fiscal tools available to governments in an energy-volatile world.

2 Literature Review

This section reviews relevant contributions to the literature to motivate the focus of this study and the methodology it employs. The first section examines evidence on fiscal responses to oil price shocks, highlighting that existing work fails to decompose the fiscal channel through which supply-side energy commodities operate. Second, the review surveys the broader empirical literature on institutions and fiscal policy, establishing the well-documented link between institutional quality and fiscal policy outcomes, discussing the measurement of institutional quality, and noting the absence of a composite index that captures the relevant economic institutions. Third, the review turns to political-economy mechanisms, namely the voracity effect, state capacity, and debt capacity, that provide the theoretical underpinnings for asymmetric fiscal reactions across institutional environments. Lastly, to motivate the study's methodology, a survey of the prominent methods for oil shock identification, institutional measurement, and state-dependent Local Projections is carried out.

2.1 Fiscal Responses to Energy Commodity Shocks

In the *Journal of Economics and Business*, Mollick and Nguyen (2011) studied how government spending and fiscal balances reacted to oil price shocks in oil-importing developing Southeast Asian economies. The authors hypothesized that oil price increases put upward pressure on government expenditure, while tax revenues decrease due to slower growth, causing budget balance deterioration. The study employed a structural vector autoregression (SVAR) for each country in the sample using oil price as an exogenous external shock by way of Cholesky decomposition. The key findings were that positive oil price shocks led to a significant and persistent increase in real government spending, causing fiscal deficits to widen as revenues lagged behind expenditure or declined.

In the same vein, a report by the International Monetary Fund (2015) presents a cross-country analysis of the fiscal consequences of commodity price swings in importing and exporting economies. In a yearly panel of advanced, emerging, and developing economies, the report analyzed the fiscal reaction functions from 1970 to 2014 to study the response of the primary balance, government expenditure, and revenue to changes in a country-specific commodity terms of trade index. The model was run separately for commodity importers and exporters, and interacted the commodity terms-of-trade with indices of fiscal rules, budget frameworks, and public financial management quality. The authors found that an increase in commodity terms-of-trade worsens the primary balance in a persistent manner for importers, though countries with stronger fiscal

frameworks managed to limit fiscal deterioration. This study provides evidence that institutional quality can mitigate the negative fiscal impact of oil price fluctuations.

Cologni and Manera (2008) report similar heterogeneity among oil-importing G-7 countries. The authors studied the effects of supply-side oil price shocks on output and prices using a SVAR model, showing that oil shocks raise government spending and reduce the primary surplus in some G-7 economies. The effect was unequal among the economies as the magnitude and sign vary across countries, yet the study did not identify which factors drove this heterogeneity. This study offers a framework for identifying supply-side oil shocks and tracing their fiscal impact, tracing the responses of government spending and the primary surplus. It also highlights the mechanisms through which supply-side oil shocks operate within the macroeconomy, and the channels through which they condition fiscal responses.

El Anshasy and Bradley (2012) extend the analysis by estimating a fiscal reaction function for oil-exporting countries and testing whether their primary balance response to oil price gaps is conditioned by institutional quality. In this study, the authors analyze 16 major oil-exporting countries from 1970-2009, building onto previous literature by implementing smooth-transition regressions to allow the fiscal response to vary with the size and sign of the oil price gap. Institutional conditioning is split beforehand into strong and weak governance based on the median of the Worldwide Governance Indicators of the sample, and then separate smooth-transition regressions are run for each group. The authors found that a positive oil price gap modestly improves the primary balance for both regimes, while strong-governance groups see significantly greater primary balance improvements after positive oil gaps than weak-governance groups. This asymmetry was more pronounced for large positive oil price gaps, as countries with more robust governance saved a higher fraction of windfalls, while institutionally weak countries spent the windfall excess. This study provides valuable methodological advances by implementing smooth-transition models that allow the fiscal response to vary continuously with the size of the oil price gap, rather than imposing a linear relationship.

These studies support the contention that institutional quality moderates fiscal responses to commodity and oil price movements in both importer and exporter economies. However, no study specifically models fiscal response heterogeneity to supply-side oil shocks in importing economies.

To construct a test that overcomes these limitations, it is first necessary to examine two foundational pillars: the broader empirical relationship between institutional quality and fiscal policy and the political-economy mechanisms that predict asymmetric expenditure-revenue behaviour.

2.2 Institutional Constraints in Fiscal Policy

Alesina, Campante, et al. (2008) study the procyclicality of fiscal policy, aiming to determine whether developing countries condition government spending on business cycles and which institutional components drive this phenomenon. The authors estimate a panel regression of 83 countries from 1960 to 2003, with the cyclical component of real government spending as the dependent variable and the main explanatory variable as the output gap interacted with institutional indicators from the ICRG and Polity IV. The study found that procyclicality is driven by corruption and weak rule of law, rather than borrowing constraints alone. Most notably, it found that the revenue side is the primary transmission channel of procyclicality, as in corrupt countries revenues rise during booms and are immediately spent. These findings establish that institutional quality is a decisive factor in how governments manage fiscal pressures that affect the revenue base.

While Alesina, Campante, et al. (2008) examine spending response to cyclical revenue swings, their core insight that weak institutions lead to a failure to mobilize or retain revenue in the face of a fiscal shock is directly relevant to the oil shock fiscal response literature. An adverse oil supply shock represents a negative income shock for oil importers, and the institutional capacity to raise off-setting revenue through windfall taxes or effective tax enforcement should, by the same logic, be concentrated in countries with strong institutions. This provides a theoretical basis for examining whether institutional quality shapes fiscal responses not only over the business cycle but also during exogenous energy supply shocks, and whether these effects operate through both spending and taxation channels.

Alesina, Campante, et al. (2008) explore the idea of cyclicity of fiscal policy with respect to commodity price cycles in a non-linear way, seeking to establish a threshold level of institutions above which fiscal policy becomes counter-cyclical. Their research leverages a panel smooth-transition regression of 132 countries to study how commodity terms-of-trade gaps impact the cyclical component of real government spending. The coefficient on the terms-of-trade gap is allowed to vary smoothly with institutional quality through a logistic transition function, so the threshold is continuous and estimated endogenously. The methodology and findings of this study reinforce the argument that institutional quality is better represented as a continuum, and that the conditioning of fiscal responses should therefore be continuous rather than binary: fiscal behavior does not switch abruptly at some institutional threshold, but evolves gradually as institutional quality improves.

2.3 Political Economy Mechanisms of Fiscal Response

Political-economy theory identifies several institutional components as central to a government's ability to tax, spend, and manage external shocks. By studying how these components shape the state's fiscal capacity and the incentives of policymakers and interest groups, this literature provides clear, testable predictions about the direction and composition of fiscal responses to price windfalls or external oil shocks.

Voracity Effect and Common-Pool Fiscal Dynamic

The voracity effect, as formalised by Tornell and Lane (1999), describes a common-pool problem in fiscal policy. When a positive revenue windfall arrives, multiple powerful interest groups compete for the surplus, driving spending up and blocking revenue-raising reforms. The result is excessive expenditure and an inability to save or increase taxes. While the original mechanism was conceived for resource-exporting countries receiving a windfall, the underlying logic applies symmetrically to oil-importing countries facing an adverse supply shock.

For an oil importer, a negative oil supply shock does not generate a fiscal windfall. Instead, it raises import costs, depresses real incomes, and shrinks the revenue base. Nonetheless, the same common-pool dynamics emerge: interest groups resist bearing the adjustment cost, demanding continued subsidies, transfers, or tax relief, while simultaneously blocking spending cuts and tax increases elsewhere. This produces sticky expenditure and stagnating or falling revenues, not because of a surplus fight, but because fragmented interests prevent the government from implementing the necessary fiscal consolidation. Alesina and Drazen (1991) capture this as a “war of attrition”, where in polarised political settings each group delays conceding, hoping others will shoulder the cost, so fiscal stabilisation is postponed.

In high-institution settings, the opposite occurs. OECD (2023) documents that during the recent energy crises, OECD countries were able to introduce or expand windfall profit taxes on fossil-fuel companies. These taxes were feasible because of strong tax administrations, broad-based compliance, and political accountability. Even though supply shocks are adverse, robust-institutionalized governments can raise revenue, thereby containing deficits and avoiding the voracity-type spending trap.

State Capacity: Legal, Fiscal, and Bureaucratic Foundations

Three dimensions of institutional quality underpin a state's capacity to manage external shocks. First, rule of law is essential for a functioning revenue system. Besley and Persson (2011) show that legal capacity—the ability of courts and enforcement agencies to uphold property rights and tax obligations—is a prerequisite for effective tax collection. Without these institutions, collection becomes arbitrary and evasion erodes the tax base. Second, control of corruption determines whether fiscal pressures

are managed through broad-based adjustments or through rent-seeking. Kleven et al. (2016) provide the micro-foundations: in corrupt environments, tax enforcement is easily undermined, so the effective tax base shrinks precisely when the state attempts to raise revenue. Third, bureaucratic quality underpins the state's ability to collect taxes and manage public spending. Gordon and Li (2009) document that in developing countries, the tax base is systematically shaped by enforcement constraints; the inability to monitor informal activity forces governments to rely on easily observable tax handles such as corporate and trade taxes. This implies that governments with weak bureaucracies cannot mobilize revenue in response to shocks regardless of the tax instruments used.

Debt Capacity and Fiscal Space

In addition to tax and expenditure channels, institutional quality also determines how much governments can borrow to smooth external shocks. Reinhart et al. (2003) explore the idea of debt intolerance thresholds, using regressions of credit ratings and risk premia on debt levels and institutional indicators. The authors argue that these thresholds are not purely economic but are deeply tied to institutional quality, historical default record, and political stability. Advanced economies can sustain high debt ratios due to their institutions and repayment track records that anchor market confidence, while emerging and developing economies lose market access at much lower debt levels. The results show that emerging economies with weak institutions and a record of defaults experience market stress at external debt ratios as low as 15–30%.

Institutional credibility is the key mechanism: a history of upholding contracts and maintaining low inflation signals to creditors that debt will be serviced, expanding fiscal space. This finding implies that in crisis scenarios, low-institution countries face borrowing constraints and are forced to keep debt low, cutting expenditure or allowing the primary balance to deteriorate.

Fiscal space, defined as the capacity to run counter-cyclical fiscal policy without endangering debt sustainability, is also a key mechanism through which government-spending and tax-rate patterns vary according to institutional environments. Aizenman et al. (2019) find that fiscal space is strongly correlated with institutional quality and financial market depth, suggesting that countries with high fiscal space, led by strong institutions and high credit ratings, pursue counter-cyclical policies that raise debt and use it as a shock-absorption tool, while low-institution countries would not be able to borrow on favourable terms and simply adjust through spending cuts.

Taken together, these theories provide a framework that maps directly onto the institutional dimensions captured by the composite index constructed in this thesis. The methodology of this study provides a threshold measure based on the theoretical

mechanisms reviewed in this section, allowing a direct test of the expenditure-deficit asymmetry predictions of the political-economy literature.

2.4 Oil Shock Identification and Measuring Institutions

Addressing the aforementioned gaps requires a methodological approach that is absent from the current literature.

Kilian (2009) shows that oil price changes can be decomposed into supply, aggregate demand, and precautionary demand shocks, all of which have different macroeconomic effects. In this study, the author uses a SVAR with sign restrictions to show that oil supply disruptions account for a modest share of oil price variation, and that their effects on output are delayed and smaller than demand-driven price increases. This finding lies at the crux of this thesis' methodological strategy: identifying an exogenous oil supply shock that is not confounded by global demand changes to estimate clean estimates of economies' fiscal responses.

A commonly used measure of oil supply shocks is the indicator developed by Känzig (2023), which is designed to capture unexpected changes in oil supply. This paper identifies oil supply news by studying high-frequency movements in oil prices around OPEC meetings and press releases. A daily shock series is extracted by embedding high-frequency surprises as external instruments in a SVAR identified with sign restrictions. The resulting measure captures news about current and future oil supply that is orthogonal to global demand, providing a well-identified supply-side oil shock. This shock series has been shown to have large and persistent effects on real oil prices and global economic activity.

Kaufmann et al. (2010) developed one of the most frequently cited institutional indices, the Worldwide Governance Indicators, which decompose institutional quality into six dimensions: Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption. Their methodology employs an unobserved components model that aggregates underlying variables from diverse sources. The authors demonstrate that while the six dimensions are highly correlated, they contain distinct information, and that a simple unweighted average of the dimensions is a transparent and robust way to form a composite index. This principle provides a foundation for constructing a composite measure of institutional quality, on the grounds that multiple institutional dimensions are needed to capture the fiscal-capacity channels identified in the political-economy literature.

2.5 State-Dependent Local Projections and Threshold Estimation

Jorda (2005) introduced Local Projections as an alternative to VAR-based impulse response estimation. Instead of iterating a VAR forward and converting estimates into impulse response functions, Local Projections estimate the response at each horizon h via a single-equation regression of the outcome variable at time $t + h$ on the shock and a set of control variables at time t .

This framework is extended by Auerbach and Gorodnichenko (2012) to allow coefficients and impulse responses to vary smoothly across a state variable. The Smooth-Transition Local Projection models the impulse response as a weighted average of two regime-specific responses, with weights given by a logistic function of the state variable. In their original application, the state is business cycle and the transition function governs how the fiscal multiplier changes as the economy shifts from expansions to recessions. This framework can be extended to any other state-dependent model, where parameters are estimated endogenously and fiscal responses transition smoothly between regimes.

Ramey and Zubairy (2018) apply the state-dependent local projection approach to fiscal policy analysis, emphasizing the importance of including a sufficient number of lags of macroeconomic variables and of the shock as controls, to absorb pre-existing dynamics and avoid omitted variable bias. Together, these studies provide the foundation for the econometric specification of this study.

3 Methodology

3.1 State-Dependent Local Projections

The empirical analysis begins with the local projection (LP) method of Jorda (2005), which estimates impulse response functions (IRFs) by running separate regressions for each horizon h rather than iterating a vector autoregression (VAR) forward. This approach offers two key advantages in the present setting. First, it is robust to dynamic misspecification, as LPs do not impose cross-equation restrictions. Second, LPs remain consistent under very weak conditions, and inference is straightforward even with persistent data (Montiel Olea and Plagborg-Møller 2021). These properties are especially valuable in a cross-country panel with substantial heterogeneity, where the true data-generating process is unlikely to be well captured by a single global VAR. Third, and of particular relevance here, the LP framework is better suited to the annual frequency of the fiscal variables: the relatively short time-series dimension means

that estimating a richly parameterized SVAR for each country is infeasible, whereas the single-equation LP can be reliably estimated with fewer observations.

3.1.1 Econometric Specification

To test whether institutional quality conditions the fiscal response to an oil supply shock, the standard LP is embedded in a state-dependent framework. Unlike the original Auerbach and Gorodnichenko (2012) applications that condition on the business cycle, the state variable here is a country’s institutional capacity. The baseline specification uses a binary threshold to split the sample into low- and high-institution regimes. For each horizon $h = 0, 1, \dots, H$ and country i in year t , the specification is:

$$\Delta_h pb_{i,t+h} = \alpha_i^h + \beta_{low}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst}_i \leq \gamma) + \beta_{high}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst}_i > \gamma) + \Gamma^h \mathbf{Z}_{it}^{base} + \varepsilon_{i,t+h} \quad (1)$$

$$\mathbf{Z}_{it}^{base} = \left(\Delta pb_{i,t-1} \quad \Delta pb_{i,t-2} \quad shock_{t-1} \quad shock_{t-2} \right)^\top$$

The dependent variable $\Delta_h pb_{i,t+h} = pb_{i,t+h} - pb_{i,t}$ is the cumulative change in the primary balance as a percentage of GDP from the year before the shock to horizon h . It measures the total fiscal adjustment over the h -year window, capturing both the immediate and the delayed responses of fiscal policy.

β_{low}^h and β_{high}^h are the cumulative impulse responses for, respectively, countries whose institutional index lies at or below the threshold γ (the low-institution regime) and those above it (the high-institution regime). Since the shock is a positive oil supply disruption, a negative β_{low}^h means countries with weak institutions allow their primary balance to deteriorate. A positive coefficient for β_{high}^h indicates that strong institutions do not incur significant deficits as a result of oil shocks.

The variable $shock_t$ is the Känzig (2023) oil supply news shock. The shock is exogenous and enters the regression directly. This is discussed in more detail in the variable construction section.

The institutional indicator $\overline{inst}_i$ is the sample average of the composite V-Dem index defined in Section 3.2.1, taken over all available years for each country. A time-invariant institutional measure serves two purposes. First, institutional quality is highly persistent, so the sample mean captures a country’s long-run institutional environment while smoothing out short-term measurement error. Second, keeping the threshold fixed over time ensures that the regime assignment is predetermined with respect to the oil shock: a country’s institutional “type” is not allowed to respond endogenously to the shock itself (Acemoglu et al. 2001). The indicator function $\mathbb{I}(\cdot)$ splits the sample into two regimes: a country belongs to the low-institution regime if its index is at or below the threshold γ , and to the high-institution regime otherwise.

The baseline specification deliberately imposes a rigid binary split to serve as a transparent and easily interpretable starting point that answers the question "are the average fiscal responses of high- and low-institution countries different?". The subsequent analysis is extended to a smooth-transition local projection, in which the institutional threshold becomes a continuous function that allows the fiscal response to evolve gradually with institutional quality. The binary model offers a baseline to anchor further specifications, while the smooth-transition model tests whether the conclusions are sensitive to the assumption of a discrete jump.

The main feature of this state-dependent design is the interaction of the oil shock with the institutional indicator. By multiplying the same exogenous shock by two mutually exclusive dummies, the specification allows the same global oil disruption to produce two different impulse responses depending on a country's institutional quality. This allows the study of fiscal response heterogeneity along the institutional spectrum, as institutions propagate that shock through the fiscal accounts of each regime-specific economy. This tests the predictions of political-economy literature: in low-institution settings, the voracity effect and weak tax capacity may translate the shock into spending pressures and revenue losses, while in high-institution settings, tax enforcement and budget discipline may instead allow offsetting revenue measures.

The term α_i^h denotes a full set of country fixed effects, which absorb all time-invariant characteristics of each country, such as geography, colonial history, legal systems, and institutional level. Since $\overline{inst}_i$ is time-invariant, the main effect of institutions cannot be separately identified from the fixed effects, so the only the interaction of institutions with the shock remains in the estimation. In practice, the fixed effects are removed by the within-transformation before OLS estimation, ensuring that the coefficients β^h are identified purely from the differential time-series response to the shock across institutional regimes.

The vector Z_{it} contains two lags of the dependent variable, i.e. the change in the primary balance, and two lags of the exogenous oil supply shock, following the approach of Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018). Including lags of the dependent variable and the shock eliminates any mechanical serial correlation induced by the overlapping projection horizons. Subsequent robustness exercises include macro-controls to absorb pre-existing cyclical dynamics that could contaminate the impulse responses.

3.1.2 Threshold γ and Grid Search

The binary institutional threshold γ is estimated by a grid search that minimizes the sum of squared residuals summed over all horizons $h = 0, \dots, 4$ (Hansen 1999). The

impulse response function is reported as the sequence $\{\beta_{low}^h\}_{h=0}^H$ and $\{\beta_{high}^h\}_{h=0}^H$. These IRFs are average responses for countries within each regime, because the institutional measure is time-invariant and the regime is fixed for a given country. All reported confidence bands are based on a non-parametric bootstrap with $n = 1000$ replications.

3.2 Robustness Controls

Control Vector Z_{it} Extension While the controls in the baseline specification absorb the most immediate cyclical dynamics, several other factors could simultaneously influence oil price and fiscal outcomes, potentially biasing the estimated impulse responses. To guard against this, the baseline model is re-estimated with an expanded set of controls.

$$\mathbf{Z}_{it}^{\text{ext}} = \begin{pmatrix} \Delta pb_{i,t-1} \\ \Delta pb_{i,t-2} \\ \text{shock}_{t-1} \\ \text{shock}_{t-2} \\ \text{gdp growth}_{i,t-1} \\ \text{inflation}_{i,t-1} \\ \text{world gdp growth}_{i,t-1} \\ \text{gpr}_{i,t-1} \\ \text{energy imports}_{i,t-1} \\ \text{oil share}_{i,t-1} \end{pmatrix}$$

The lags of GDP growth and CPI inflation safeguard against the risk that estimated impulse responses capture ordinary business-cycle fluctuations rather than the effect of the oil supply shocks. Without these lags, a positive oil supply shock that coincides with a cyclical downturn would incorrectly attribute the fiscal deterioration to the oil shock.

Furthermore, the dependent variable is the primary balance as a share of nominal GDP: $\frac{PB_{it}}{GDP_{it}}$. An oil supply shock that raises the general price level through cost-push inflation would mechanically decrease the dependent variable, even if real fiscal policy has not changed. In a similar vein, a positive oil supply shock that coincides with a cyclical downturn would incorrectly attribute the subsequent fiscal deterioration to the oil shock, when in fact it reflects an ongoing recession.

World GDP growth is also added to capture global demand trends, as these could raise oil prices and increase fiscal revenues in importing countries. The Geopolitical Risk Index helps control for periods of heightened geopolitical uncertainty. Military conflicts, terrorism, and international tension may affect both energy markets and government spending decisions, thus controlling for this factor absorbs the fiscal responses

driven by security-related spending.

Energy intensity, proxied by the countries share of primary energy consumption from oil and energy imports as a share of GDP, determines how strongly an oil price change feeds through to production costs and, consequently, to tax revenues and subsidies. A critical concern is that the institutional threshold could reflect the connection between institutions and energy-intensiveness, so this relationship is also tested in a separate model.

The extension parameters are kept at a single lag to keep the model as parsimonious as possible. Expanding the lag length for every variable would decrease the degrees of freedom, inflate standard errors, and reduce the power of the bootstrap inference.

Additive Interaction with Energy Imports To disentangle the interaction with energy imports in a robust manner, the baseline binary-threshold LP is extended with an additive interaction between the oil supply shock and the energy-import-to-GDP ratio. The specification allows the impulse response to vary linearly with import dependence within each institutional regime, without altering the threshold itself.

$$\begin{aligned}
 \Delta_h pb_{i,t+h} = & \alpha_i^h + \beta_{\text{low}}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst}_i \leq \gamma) \\
 & + \beta_{\text{high}}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst}_i > \gamma) \\
 & + \Psi_{\text{low}}^h \cdot shock_t \cdot (energy_imports_{it} - \overline{energy_imports}) \cdot \mathbb{I}(\overline{inst}_i \leq \gamma) \\
 & + \Psi_{\text{high}}^h \cdot shock_t \cdot (energy_imports_{it} - \overline{energy_imports}) \cdot \mathbb{I}(\overline{inst}_i > \gamma) \\
 & + \Gamma^h \mathbf{Z}_{it} + \varepsilon_{i,t+h}
 \end{aligned} \tag{2}$$

The total effect for a country with imports SD above the mean is then:

$$\begin{aligned}
 \text{Total IRF}_{\text{low}} &= \beta_{\text{low}}^h + \Psi_{\text{low}}^h \cdot SD(energy_imports), \\
 \text{Total IRF}_{\text{high}} &= \beta_{\text{high}}^h + \Psi_{\text{high}}^h \cdot SD(energy_imports).
 \end{aligned}$$

where $energy_imports_{it}$ is demeaned by the sample mean $\overline{energy_imports}$. The coefficients Ψ_{low}^h and Ψ_{high}^h capture the additional effect of a higher import dependence on the fiscal response, over and above the institutional regime average. The key question is whether the institutional gap persists after the inclusion of these terms.

3.2.1 Time Trend and Real Fiscal Variables

A final robustness check tests an important concern with the long-horizon fiscal panel. The oil-shock literature reports that the macroeconomic impact of oil supply shocks may have declined over time. Baumeister and Peersman (2013) found that oil supply

disruption on global output fell after the 1980s. The presence of this time trend in the latter half of the sample for high-institution countries could produce a spurious institutional gap, where the difference in fiscal responses could be driven by time rather than institutional capacity.

A natural solution would be to include time fixed effects, but these would absorb all variation that is common across countries in a given year. A linear time trend serves as a compromise, removing the the smooth decline in shock impact that might correlate with institutional quality, while preserving the common annual variation.

$$\Delta_h pb_{i,t+h} = \alpha_i^h + \beta_{\text{low}}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst_i} \leq \gamma) + \beta_{\text{high}}^h \cdot shock_t \cdot \mathbb{I}(\overline{inst_i} > \gamma) + \theta^h \cdot t + \Gamma^h \mathbf{Z}_{it}^{\text{ext}} + \varepsilon_{i,t+h} \quad (3)$$

where t is a linear year index. If the institutional gap in the IRFs remains similar after the inclusion of the time trend, it is unlikely to be a construct in the fiscal transmission of oil shocks.

To rule out the possibility that the institutional asymmetry is driven by nominal fluctuations, the primary balance is converted into real terms using the GDP deflator and the baseline local projection is re-estimated with this real measure as the dependent variable.

The dependent variable is the cumulative change in the inverse hyperbolic sine of the real primary balance. This transformation is applied because the IHS approximates the natural logarithm and is linear near zero, allowing the impulse-response functions to be interpreted as approximate percentage changes in the real primary balance (Bellemare and Wichman 2020).

$$\begin{aligned} \Delta_h ihs_pb_{i,t+h} &= arsinh(real_pb_{i,t+h}) - arsinh(real_pb_{i,t}) \\ \Delta_h ihs_pb_{i,t+h} &= \alpha_i^h \\ &\quad + \beta_{\text{low}}^h shock_t \cdot \mathbb{I}(\overline{inst_i} \leq \gamma_{\text{real}}) \\ &\quad + \beta_{\text{high}}^h shock_t \cdot \mathbb{I}(\overline{inst_i} > \gamma_{\text{real}}) \\ &\quad + \Gamma^h \mathbf{Z}_{it}^{\text{base}} + \varepsilon_{i,t+h} \end{aligned} \quad (4)$$

All other components of $\mathbf{Z}_{it}^{\text{base}}$ remain identical to the baseline specification.

3.3 Smooth-Transition Local Projections

The baseline specification's rigid binary split discards all within-regime variation in institutional quality and does not allow a country's institutional capacity to evolve over

time. The review of the political-economy literature suggests that institutional quality operates on a spectrum, and that countries experience institutional change over time, features that time-invariant measures cannot capture. To address these concerns, the binary threshold is replaced by a Smooth-Transition Local Projection (STLP), following Auerbach and Gorodnichenko (2012). In this specification, the fiscal response to the oil shock is not forced into two discrete boxes but is allowed to vary continuously with the level of institutional quality, and the institutional index is permitted to fluctuate annually.

$$\begin{aligned}\Delta_h pb_{i,t+h} = & \alpha_i^h + \beta_{low}^h \cdot shock_t \cdot (1 - F(z_{it})) \\ & + \beta_{high}^h \cdot shock_t \cdot F(z_{it}) \\ & + \Gamma^h \mathbf{Z}_{it} + \varepsilon_{i,t+h},\end{aligned}\tag{5}$$

where z_{it} denotes the annual composite institutional index, and $F(z)$ is the logistic transition function:

$$F(z_{it}) = \frac{1}{1 + \exp(-\lambda(z_{it} - c))}.$$

The parameter c is the center of the transition: when $z_{it} = c$, $F = 0.5$ and the impulse response is an equal mixture of the two extreme regimes. $\lambda > 0$ controls the speed of transition, where a large λ makes $F(z)$ approach a step function similar to the binary baseline.

β_{low}^h is the cumulative IRF for a country with extremely weak institutions, and β_{high}^h is the cumulative IRF for a country with extremely strong institutions. For any country-year observation with institutional score z_{it} , the estimated impulse response is a weighted average $\beta_{low}^h \cdot (1 - F(z_{it})) + \beta_{high}^h \cdot F(z_{it})$.

The parameters λ and c are unknown and are estimated jointly by a two-dimensional grid search, using the same methodology as in the baseline model.

The STLP provides a direct answer to the question: How does the fiscal response vary across the entire spectrum of institutional quality, without imposing an arbitrary breakpoint? By replacing the discrete $\mathbb{I}(\cdot)$ indicator with a smooth function, the model allows the data to determine the shape of the transition.

The robustness of the STLP is assessed by re-estimating the model with the extended control vector $\mathbf{Z}_{it}^{\text{ext}}$, exactly as in the robustness checks for the baseline LP.

3.4 Fiscal Decomposition

The baseline model and the smooth-transition extension estimate the impulse response of the primary balance, a summary fiscal indicator. As a result, a statistically significant difference between β_{low}^h and β_{high}^h would confirm that institutions condition

the overall fiscal outcome, but would not reveal how the adjustment takes place. Political-economy theory makes precise predictions about which margin should adjust in each institutional setting, and testing those predictions requires a decomposition of the primary balance into its expenditure, revenue, and debt components.

3.4.1 Estimation and Regime Definition

The decomposition is implemented by re-estimating the baseline specification with the dependent variable replaced by:

$$\Delta_h Y_{i,t+h} = \alpha_i^h + \beta_{\text{low}}^h \cdot \text{shock}_t \cdot \mathbb{I}(\overline{\text{inst}_i} \leq \gamma) + \beta_{\text{high}}^h \cdot \text{shock}_t \cdot \mathbb{I}(\overline{\text{inst}_i} > \gamma) + \Gamma^h \mathbf{Z}_{it}^{\text{base}} + \varepsilon_{i,t+h}, \quad (6)$$

$$Y \in \{\text{expenditure, government_rev, tax_rev, debt}\}$$

The institutional threshold γ is held constant at the value estimated from the primary-balance baseline. This ensures that the same regime definition applies to all four fiscal variables, so that differences in their impulse responses reflect differential transmission of the shock to the three channels rather than changes in the sample split.

3.4.2 Political Economy Hypotheses

The political-economy literature offers clear predictions on how institutional quality should shape the fiscal transmission of an oil supply shock, and the decomposition of the primary balance into expenditure, tax revenue, and debt allows each prediction to be tested separately.

- **Government expenditure** isolates the voracity effect (Tornell and Lane 1999): in low-institution countries, competing interest groups resist spending cuts and may push for compensatory spending, so expenditure should rise significantly; in high-institution countries it should remain flat or fall.
- **Tax revenue** captures the state-capacity channel (Besley and Persson 2011; Kleven et al. 2016): high-institution countries can raise offsetting revenue (e.g. windfall taxes), whereas low-institution countries, hampered by weak enforcement and large informal sectors, should see tax revenues stagnate or decline.
- **Public debt** reflects the fiscal-space and debt-intolerance mechanisms (Reinhart et al. 2003; Aizenman et al. 2019): high-institution countries are expected to let debt rise temporarily as a shock absorber; low-institution countries, facing tighter

borrowing constraints, should show little to no debt accumulation, or even a decline.

These predictions are tested by comparing the horizon-specific impulse responses for each fiscal component across the two institutional regimes, using the same threshold γ and bootstrap-based confidence bands as in the baseline primary-balance specification.

3.4.3 Robustness Controls

All fiscal variables in the panel are expressed as ratios of nominal aggregates to nominal GDP (% of GDP). To verify that the institutional asymmetry reflects real fiscal adjustment rather than nominal dynamics, each fiscal variable is converted to constant price units. Given a fiscal ratio R_{it}^X for variable X (expenditure, revenue, debt, or the primary balance), the real variables are computed as:

$$\text{Real } X_{it} = \frac{R_{it}^X}{100} \times \text{real_gdp}_{it},$$

Real GDP in constant 2015 US dollars (NY.GDP.MKTP.KD) is used to convert the fiscal ratios into real terms. This transformation preserves the cross-country comparability of the original ratios while removing the mechanical effect of inflation on the denominator, as done in the robustness section for the baseline model.

Expenditure, revenue, and debt enter the LP as natural logarithms, and the dependent variable in the local projections becomes the cumulative change in the log of the real fiscal aggregate $\Delta_h \ln(\text{Real } X_{i,t+h})$. This estimates the approximate percentage change of the fiscal variable over the horizon, allowing the impulse-response functions to be interpreted as elasticities.

3.5 Institutional Decomposition

While the baseline binary threshold and the smooth-transition specifications aims to determine whether the composite institutional index moderates the fiscal response to oil supply shocks, they do not reveal which dimensions of institutional quality are responsible for the aggregate result. Different political-economy mechanisms operate through distinct institutional features. Bureaucratic quality lies at the center of the voracity effect, property rights and transparent laws are central to fiscal capacity, and electoral democracy and control of corruption underpins the political accountability channel. To isolate the contribution of each institutional pillar, the composite index is disaggregated and the analysis is repeated using each standardized sub-index as the state variable.

For each sub-index $j \in \{\text{control of corruption, rule of law, property rights, transparent laws, bureaucratic quality, democracy}\}$ and each fiscal outcome $Y \in \{\text{primary balance, tax revenue, expenditure, debt}\}$, a smooth-transition local projection is estimated:

$$\begin{aligned}\Delta_h Y_{i,t+h} = & \alpha_i^h + \beta_{F=0}^h \cdot shock_t \cdot (1 - F^j(z_{it}^j)) \\ & + \beta_{F=1}^h \cdot shock_t \cdot F^j(z_{it}^j) \\ & + \Gamma^h \mathbf{Z}_{it}^{\text{base}} + \varepsilon_{i,t+h}\end{aligned}\tag{7}$$

The control set $\mathbf{Z}_{it}^{\text{base}}$ is identical to the baseline specification (two lags of the dependent variable and two lags of the oil shock). All coefficients are indexed by the horizon h .

For each pair (j, Y) , the parameters c^j and λ^j are estimated by a two-dimensional grid search that minimises the sum of squared residuals over horizons $h = 0, \dots, 4$, exactly as in the aggregate smooth-transition model.

To obtain a comparable quantitative measure of the institutional gradient produced by each sub-index, the impulse response is evaluated at two benchmark values of the state variable: the 10th percentile (**low-institution** scenario) and the 90th percentile (**high-institution** scenario) of the sample distribution of z_{it}^j . The cumulative impulse response at horizon h for a given scenario is constructed as a weighted average of the extreme-regime coefficients:

$$\text{IRF}_{\text{low}}^h = \beta_{F=0}^h \cdot (1 - F^j(p_{10})) + \beta_{F=1}^h \cdot F^j(p_{10}), \quad \text{IRF}_{\text{high}}^h = \beta_{F=0}^h \cdot (1 - F^j(p_{90})) + \beta_{F=1}^h \cdot F^j(p_{90}).$$

The moderation effect at horizon 4, $\Delta^4 = \text{IRF}_{\text{high}}^4 - \text{IRF}_{\text{low}}^4$, captures the additional cumulative fiscal adjustment that occurs when a country moves from the lower to the upper end of the institutional spectrum. A positive Δ^4 indicates that stronger institutions are associated with a larger improvement in the primary balance, a greater increase in tax revenue or debt, or a smaller cut in expenditure, depending on the outcome variable. By ranking the 24 institution–fiscal pairs according to $|\Delta^4|$, the decomposition reveals which institutional dimension matters most for each fiscal channel.

This decomposition aims to provide a way to test the theoretical mechanisms discussed in the political-economy literature. Furthermore, it aims to identify which fiscal instrument among debt, expenditure, revenue, or the primary balance is most sensitive to institutional variation, thereby complementing the fiscal decomposition.

4 Variable Construction

4.1 Institutional Composite Index

The preceding literature review emphasized the importance of capturing all the institutional dimensions through which exogenous shocks transmit to fiscal outcomes. For oil-importing economies, adverse oil supply shocks act as negative supply-side cost shocks, not as cyclical booms or busts. The relevant institutional question is therefore one of institutional capacity—the ability to raise offsetting tax revenue, control spending, and manage debt—rather than cyclical discipline alone. Political-economy theory identified three specific dimensions that govern that capacity: rule of law, control of corruption, and bureaucratic quality. Each of these is measured using a set of expert-coded indicators from the V-Dem (Varieties of Democracy) dataset (Coppedge et al. 2026), which provides annual institutional indicators for over 200 countries. The construction of every sub-component is described below, and all variables are used in the most recent v14 release.

Rule of Law and Legal Capacity Rule of law ensures that tax claims are enforceable and that fiscal contracts, with both creditors and citizens, are credible. Three V-Dem indicators capture distinct aspects of this dimension

- **Rule of Law** (v2x_rule): This is V-Dem’s composite index measuring the extent to which laws are transparently, independently, predictably, impartially, and equally enforced, and the degree to which government actions comply with the law. It is an interval variable ranging from 0 (no rule of law) to 1 (full rule of law).
- **Property Rights** (simple average of v2clprptym and v2clprptyw): Property rights are a fundamental component of legal capacity, as without secure private property, tax obligations cannot be credibly enforced because the state cannot protect the assets on which taxes are levied. The two variables measure, on a 0–5 ordinal scale, the "extent to which men (v2clprptym) and women (v2clprptyw) enjoy the right to acquire, possess, inherit, and sell private property, including land" (Coppedge et al. 2026). The questions explicitly ask about rights, not actual ownership, and consider limits coming from the state, customary law, or social norms. The two gender-specific measures are averaged to obtain a single property rights score for each country-year.
- **Transparent laws** (v2cltrnslw): captures whether the laws of the land are clear, well publicized, and stable, and whether the legislative process provides opportu-

nities for public input. Transparent laws are a necessary condition for the rule of law, as if laws are unclear or arbitrarily changed, tax obligations become unpredictable and enforcement is impossible. The variable is measured on an ordinal scale, with higher values indicating greater transparency.

Control of Corruption Corruption is one of the most prevalent hindrances in developing nations, and they determine whether fiscal pressures are managed through broad adjustments or rent-seeking. Political-economy theory, specifically the voracity effect of Tornell and Lane (1999) predicts that where corruption is pervasive, interest groups compete for public resources, driving up spending and blocking revenue-raising measures.

- **Inverted Political Corruption Index** ($1 - v2x_corr$): measures how pervasive political corruption is, on an interval scale from 0 (least corrupt) to 1 (most corrupt). To make the direction consistent with the other indicators, where higher values denote better institutional quality, the complement $1 - v2x_corr$ is taken. The resulting variable is an index of control of corruption, where higher values indicate cleaner governance.

Bureaucratic Quality and State Capacity Bureaucratic quality underpins the state's capacity to conduct fiscal policy. Collecting taxes and manage public spending without civil service able to enforce tax laws and compliance results in an inability to carry out proper fiscal responses. These two variables measure, on a 0–4 ordinal scale, how routinely members of the executive (the head of state, the head of government, and cabinet ministers) or their agents engage in bribe-taking and embezzlement.

- **Executive Bribery** ($v2exbribe$): measures how often executive power agents grant favours for bribes or material rewards (0 = routine and expected, 4 = never, or hardly ever)
- **Embezzlement** ($v2exembezz$): measures how often executive power members steal, embezzle or misappropriate national resources of funds (0 = constantly, 4 = never, or hardly ever)

The two scores are averaged to form a bureaucratic probity index that captures the integrity of the highest level of public administration.

Accountability The extent to which citizens can hold the executive and legislative powers accountable has also been shown to condition fiscal discipline (Persson and Tabellini 2003).

- **Democracy** (v2x_polyarchy): measures the achievement of core democratic principles (free and fair elections, freedom of expression, associational autonomy, and inclusive citizenship) on an interval scale from 0 to 1. Including this dimension ensures that the composite index reflects the electoral incentives that may restrain opportunistic fiscal behavior.

All sub-indices are first standardized to have zero mean and unit standard deviation over the full cross-section of countries in each year, effectively removing the differences in scale and ensures that each dimension contributes equally to the final measure. The standardised sub-indices are then combined via a simple unweighted average, following the argument set forth by Kaufmann et al. (2010) that averaging correlated institutional dimensions yields a transparent and robust composite. The resulting index is a continuous score where higher values denote stronger institutions.

Table 1: Correlation of each sub-index with the institutional composite index

Institutional indicator	Correlation with composite
Rule of law	0.979
Transparent laws	0.957
Democracy	0.930
Bureaucratic quality	0.912
Control of corruption	0.885
Property rights	0.769

These results show the high correlation of the individual institutional variables and the composite index, making it an apt instrument to proxy each economies' total institutional quality. The ordering also matches expectations, as advanced economies with long histories of stable legal and bureaucratic institutions lay on the top of the distribution, while autocracies and fragile states have significantly negative values.

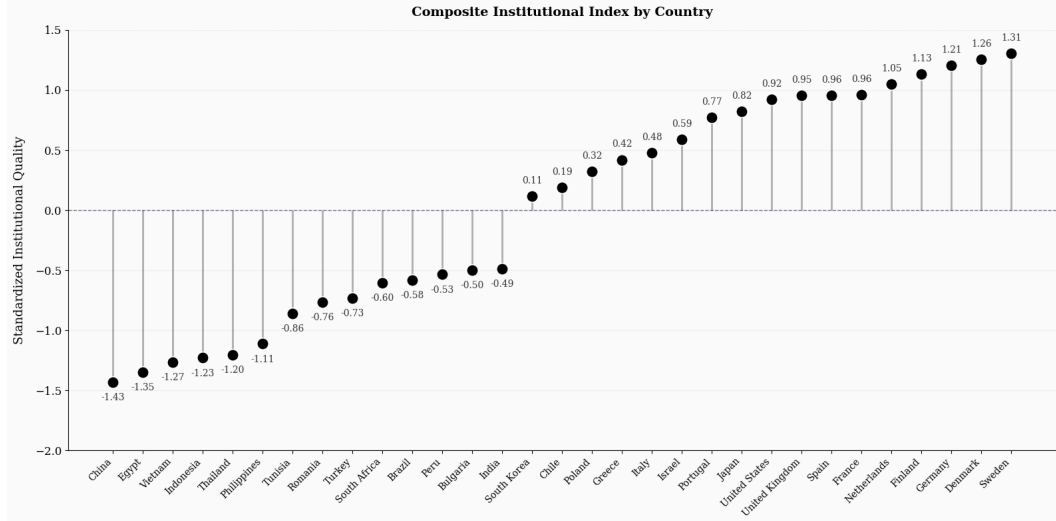


Figure 1: Country-Specific Composite Institutional Quality

4.2 Oil Supply News Shock

The oil supply shock is the Känzig (2023) oil supply news series, constructed from high-frequency movements in oil futures prices around OPEC press releases and production announcements. Since the variation is extracted from a narrow window around pre-scheduled OPEC events and is filtered through a sign-restricted SVAR that removes global demand and precautionary demand components, the resulting series captures unanticipated news about current and future oil supply that is orthogonal to aggregate demand.

The raw series is available at the monthly frequency. To match the annual fiscal panel, the monthly values are summed within each calendar year. Summation preserves the total intensity of supply news that arrives over the year and follows the logic of other narrative shock series that accumulate high-frequency surprises to a lower frequency (Ramey and Zubairy 2018).

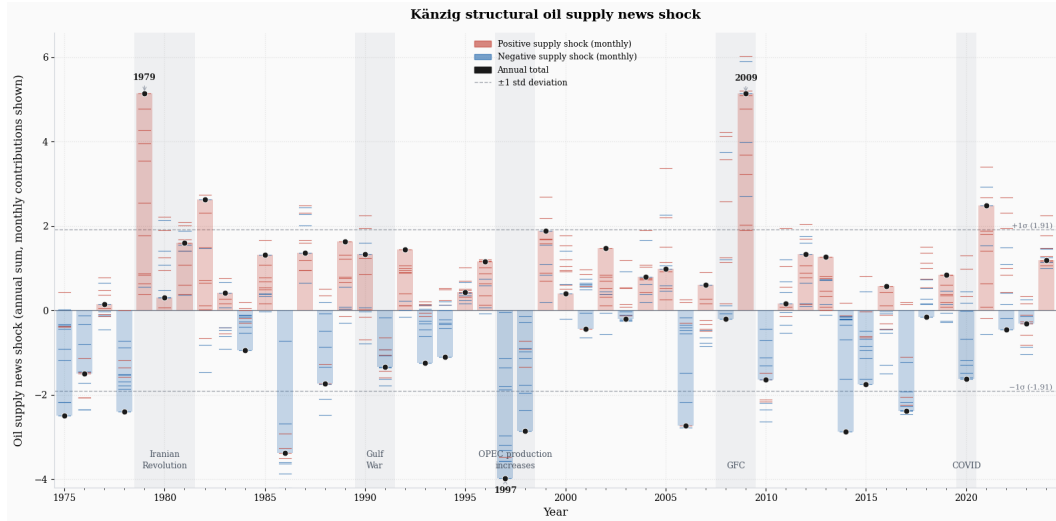


Figure 2: Annual Känzig oil supply news shock (monthly sum)

The orthogonality of the Känzig shock to aggregate demand makes it a cleaner measure than raw oil price changes, which conflate supply disruptions, global expansions, and other demand-side dynamics. The figure below illustrates this: the Brent crude price does not always track the oil supply news shock, confirming that the two series convey different information.

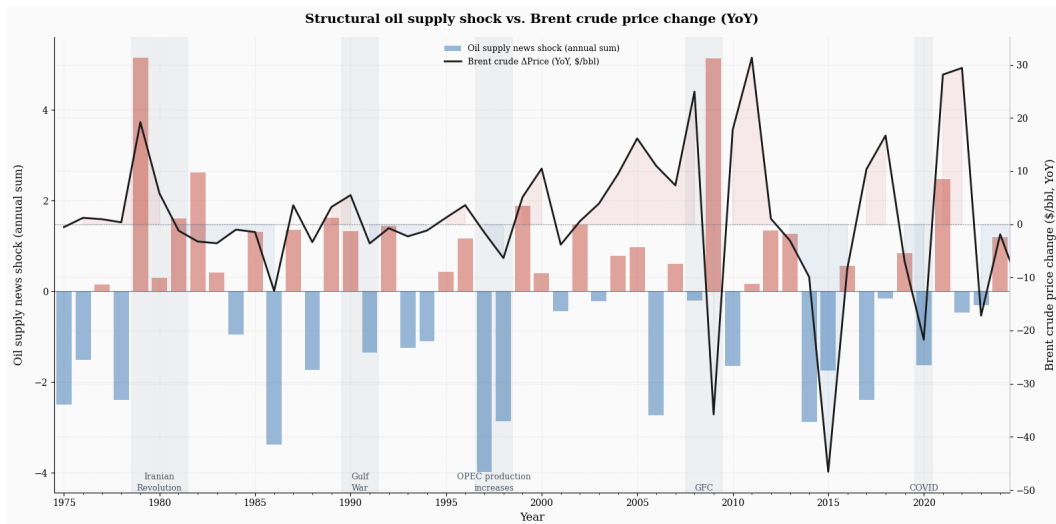


Figure 3: Supply-Side Oil Shock vs. YoY Brent Crude Price Change

The two series diverge substantially in several episodes, underscoring the distinct nature of supply news. During the mid-2000s commodity boom, Brent prices rose sharply on the back of strong global demand, while the Känzig shock remained muted. Conversely, some supply events triggered a sharp Känzig shock even though oil prices moved only modestly. Three historical episodes are particularly revealing. The 1979

Iranian Revolution caused a severe supply disruption that spiked both the Känzig shock and the Brent price, a classic supply-side event.

In 1997, favorable OPEC production increases generated a large negative supply news shock, isolating the supply component that would otherwise have been buried in the oil price decline driven by the Asian financial crisis. The most striking divergence occurs in 2009: the global financial crisis led OPEC to announce deep production cuts, producing a very positive supply news shock, while the simultaneous collapse in world demand caused Brent prices to fall sharply. The two series move in exactly opposite directions, demonstrating that the Känzig shock captures supply-side news, not demand-driven price fluctuations.

5 Data

The analysis covers 31 countries over the period 1974–2024, yielding an unbalanced panel of approximately 1,558 country-year observations. The sample is intentionally heterogeneous, combining 15 advanced economies and 16 emerging markets and developing economies as classified by the IMF World Economic Outlook. This diversity allows the analysis to assess whether fiscal responses to oil supply shocks differ systematically across institutional quality levels and stages of development.

Country Sample In this setup, all countries are net energy importers, or have been for the majority of the sample period. Notable exceptions include the United States, which was a net energy importer until 2019. This time-varying energy status does not invalidate the implications of energy shocks on fiscal policy, as the United States remains reliant on crude oil imports because its domestic production is mainly of "light" or "sweet" crude, while the majority of its refineries were built for "heavy" or "sour" crude (Communications 2025). For example, in 2024 the United States produced over 13 million barrels of oil per day while importing over 6 million barrels per day (Jacobson 2026). Indonesia is similarly retained: although classified as a net oil exporter during parts of the sample period, its domestic refineries require imported crude oil to meet domestic demand, leaving the country exposed to global oil supply shocks (U.S. Energy Information Administration 2023).

Table 2: Country Sample and Data Coverage

Country	Class	Sample	T	Primary Balance	Gross Debt	Tax/GDP
<i>Advanced Economies</i>						
Denmark	AE	1974–2024	51	Full	Full	Full
Finland	AE	1974–2024	51	Full	Full	Full
France	AE	1974–2024	51	Full	From 1978	Full
Germany	AE	1974–2024	51	Full	Full	Full
Greece	AE	1974–2024	51	Full	Full	Full
Israel	AE	1974–2024	51	From 1980 (partial)	From 1983	Full
Italy	AE	1974–2024	51	Full	Full	From 1973
Japan	AE	1974–2024	51	Full	Full	Sparse ^a
Netherlands	AE	1974–2024	51	Full	Full	From 1973
Portugal	AE	1974–2024	51	Full	Full	From 1973
South Korea	AE	1974–2024	51	Full	From 1990	Full
Spain	AE	1974–2024	51	Full	From 1980	Full
Sweden	AE	1974–2024	51	Full	Full	Full
United Kingdom	AE	1974–2024	51	Full	Full	Full
United States	AE	1974–2019	46	Full (to 2019)	Full (to 2019)	Full (to 2019)
<i>Emerging Market and Developing Economies</i>						
Bulgaria	EM	1974–2024	51	From 1995	From 1997	From 1990
Chile	EM	1974–2024	51	Full	From 1990	Full
China	EM	1974–2024	51	From 1989	From 1995	From 2005
India	EM	1974–2024	51	Full	From 1991	1974–2022
Indonesia	EM	1974–2024	51	From 1972	From 2000	1972–2009
Peru	EM	1974–2024	51	Full	From 1999	1972–2021
Philippines	EM	1974–2024	51	Full	Full	From 1990
Poland	EM	1974–2024	51	From 1995	From 1990	From 1994
Romania	EM	1974–2024	51	From 1985	From 1995	From 1981
South Africa	EM	1974–2024	51	From 1974 (partial)	Missing	Full
Thailand	EM	1974–2024	51	Full	From 2005	Full
Tunisia	EM	1974–2024	51	From 1991	Missing	1972–2012
Turkey	EM	1974–2024	51	Full	From 2000	Full
Vietnam	EM	1974–2024	51	From 1998	From 2000	Missing

Notes: AE = Advanced Economy; EM = Emerging Market and Developing Economy (IMF WEO). “Full” = available entire sample. Japan’s tax/GDP (WB) covers 1991–1993 only and is treated as missing. South Africa and Tunisia lack gross debt data.

5.1 Fiscal Variables

The fiscal data draw primarily on the IMF Historical Public Finance Database ((International Monetary Fund 2024a)), which provides long-run annual series for general government accounts covering the period 1800–2024. This is supplemented by the IMF

World Economic Outlook ((International Monetary Fund 2024b)) for more recent observations where the HPF lacks publication.

Primary balance. The central fiscal outcome variable is the general government primary net lending/borrowing balance, expressed as a share of GDP. This is defined as total revenue minus total expenditure net of interest payments, and is sourced from the International Monetary Fund (2024a), with International Monetary Fund (2024b) supplements for recent years. The raw file spans 1800-2024, and the effective sample for most countries runs from 1974 to 2024.

General government expenditure and revenue. Total general government expenditure (as a share of GDP) is drawn from the International Monetary Fund (2024a). The expenditure series follows the IMF convention of including interest payments, so that subtracting interest from expenditure recovers the primary balance. Expenditure coverage for Vietnam and Tunisia coverage begins in 1991 and 1998, respectively.

Tax revenue. Tax revenue as a share of GDP is sourced from the World Bank World Development Indicators (WDI), indicator `GC.TAX.TOTL.GD.ZS`. Coverage begins in the early 1970s for most advanced economies but is sparser for emerging markets. Brazil and Vietnam are entirely absent, and Japan’s data cover only 1991-1993 and are treated as missing throughout. This variable serves as an alternative dependent variable to assess the composition of fiscal adjustment. A companion real tax revenue series is constructed by deflating the current local-currency equivalent (`GC.TAX.TOTL.CN`) by the GDP deflator (`NY.GDP.DEFL.ZS`), providing a measure of tax collection volume that is independent of GDP movements.

General government revenue. In addition to tax revenue, the full range of the revenue side is captured by general government revenue (% of GDP), sourced from the International Monetary Fund (2024a). This variable includes all current and capital receipts (taxes, social contributions, grants, and other revenues) and is therefore broader than the tax-revenue series. Examining both measures allows the decomposition to distinguish between responses driven by deliberate tax policy and those reflecting the wider revenue base.

Gross government debt. Gross general government debt as a share of GDP is drawn from the International Monetary Fund (2024a). The raw file spans 1950-2024, and coverage begins later for many emerging markets. South Africa and Tunisia are

entirely absent. These two countries are therefore excluded from all debt-related specifications.

5.2 Macroeconomic Variables

The macroeconomic controls are sourced almost entirely from the World Bank WDI, which provides consistent annual series covering the period 1960-2025.

Real GDP growth and level. GDP growth (annual %, indicator `NY.GDP.MKTP.KD.ZG`) is the annual real GDP growth. A constant-USD GDP level (`NY.GDP.MKTP.KD`, base year 2015) is used to construct the world GDP growth aggregate as a GDP-weighted mean across countries. Both series are available from 1960; Brazil ends in 2006.

CPI inflation. Consumer price inflation is computed as the annual percentage change in the CPI index (`FP.CPI.TOTL`, base year 2010), sourced from the World Bank WDI (International Monetary Fund 2024b). Coverage runs from 1960 for most countries: Brazil is available from 1981, Romania from 1991, and Vietnam from 1996.

World GDP growth. The World Bank WDI "World" aggregate for GDP growth is included as an annual scalar broadcast to all countries (International Monetary Fund 2024b) (code `WLD`), controlling for global cyclical conditions that affect all sample members simultaneously and could otherwise confound the identification of oil-shock effects.

Brent Crude Oil Price. The level of the Brent crude oil price (USD per barrel) is drawn from the World Bank Pink Sheet commodity price data (World Bank 2024), taking its monthly prices and aggregating to annual averages.

GDP Deflator The GDP deflator (`NY.GDP.DEFL.ZS`) from the World Bank WDI is used solely as a deflator to construct real tax revenue and does not enter directly as a dependent or control variable in any specification.

Energy Exposure Variables Two country-level variables are used to measure structural exposure to external energy price shocks and to construct the heterogeneous-effects specifications.

Energy Imports The share of energy imports in total energy use (EG.IMP.CON.S.ZS) is sourced from the World Bank WDI. Coverage runs from 1990 to 2023, and the series is unavailable before 1990 for any country in the sample. This limits the regressions employing this variable to the post-1990 sub-period, which is why it is only employed as a robustness control. A higher value indicates greater dependence on imported energy and hence stronger direct pass-through from global oil prices to domestic energy costs.

Oil Share of Primary Energy. The share of oil in total primary energy consumption is sourced Ritchie et al. (2023). Annual coverage begins around 1970 for most countries, except South Africa, and Tunisia, who have partial gaps.

Geopolitical Risk Country-level geopolitical risk is measured using the index of Caldara and Iacoviello (2022), constructed from automated text searches in major international newspapers (series GPRHC_IS03). Monthly observations are aggregated to annual means to match the annual frequency of the other variables. Country-specific series are available for approximately two-thirds of the sample, with Bulgaria, Hungary and Poland lacking coverage for some years. To account for this, the EU-country average is imputed to fill the missing variables.

6 Results

The baseline State-Dependent Local Projection resulted in a stark contrast between the impulse response coefficients of low and high institution countries. The threshold γ of 0.21 split the sample into the following regimes:

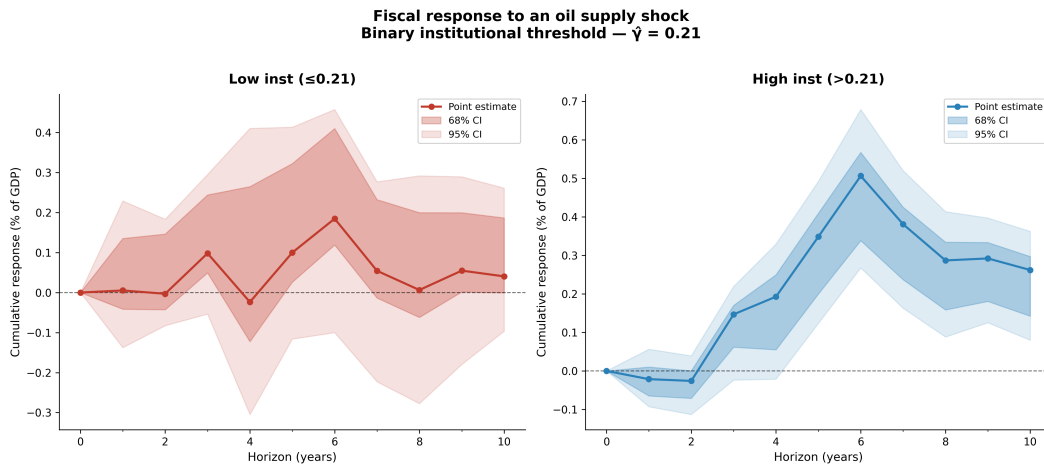


Figure 4: Baseline Local Projection Model

This division makes intuitive sense, as most developed nations, including all European countries, are in the high-institution threshold, and most global south countries fall in the low-institution regime classification. This provides evidence that the composite institutional measure correctly captured the elements in each economy responsible for fiscal response vis-a-vis oil supply shocks, as the threshold found managed to reconcile broad institutional quality of each country and the asymmetry in their fiscal responses.

This pattern suggests that high-institution countries used the shock as an opportunity to implement lasting fiscal adjustments, such as tax reforms, expenditure restraint, or strengthened budgetary frameworks. These measures could have permanently raised the primary surplus, leading to the pattern revealed by the IRFs. This response is consistent with the view that institutional quality enables disciplined, forward-looking fiscal responses, whereas weak institutions produce ad-hoc or absent reactions.

It must be noted that these responses are the averages of each regime, and there was notable exceptions within each regime, as reported by the individual IRFs in the appendix. The most notable cases were Vietnam, China and Thailand, who recorded significant and persistent improvements in their primary balances following adverse oil supply shocks despite being classified as low-institution countries.

In highly centralized political systems, the executive power faces fewer veto players and can implement rapid fiscal adjustments, cutting energy subsidies, raising administered fuel prices, or extracting additional revenues from state-owned energy companies without facing the resistance that is present in other low-institution settings (Alesina and Drazen 1991). Moreover, bureaucratic capacity in revenue collection and economic management in these countries could be substantial even when democratic accountability scores are low, a pattern well documented in East Asian developmental states (Evans 1995; Doner 2009). This pattern highlights a limitation of aggregating multiple institutional dimensions into a single score and suggests that future extensions could explore whether a narrower measure of bureaucratic quality better predicts these fiscal responses.

6.1 Robustness Controls

6.1.1 Extended Controls

The impulse responses obtained with the extended control set are consistent in direction, magnitude, and statistical significance with those from the baseline specification.

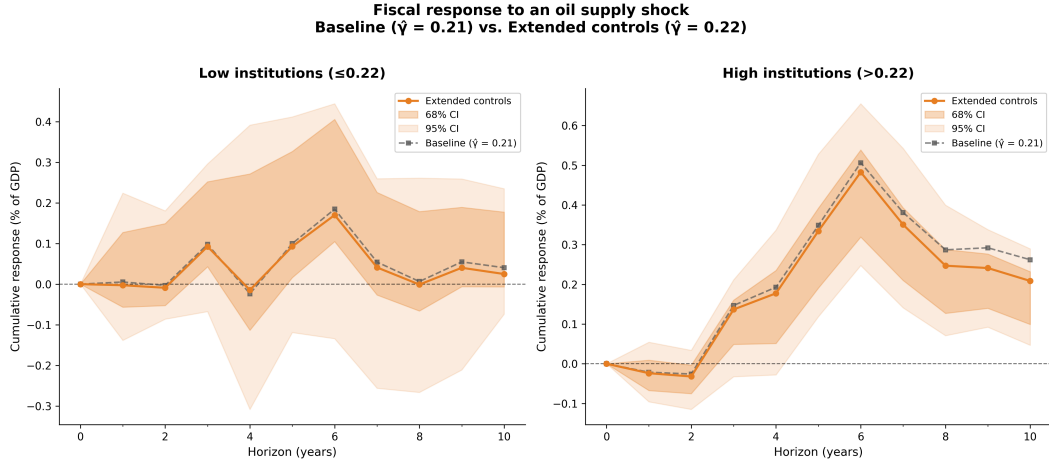


Figure 5: Extended Controls Local Projection

This result is informative in various regards. For one, the fact that controlling for world GDP growth leaves the IRFs unchanged indicates that the oil supply news shock is not inadvertently capturing common global demand expansions that could simultaneously lift fiscal revenues in high-institution countries. This supports the robustness of the oil-supply shock used, and the specification as a whole. Furthermore, the inclusion of the Geopolitical Risk Index suggests that the fiscal response is driven by the oil supply shocks itself rather than by geopolitical uncertainty. The inclusion of direct measures of energy exposure (imports and consumption) also do not erode the institutional gap, suggesting that the threshold is not proxying for how intensively a country relies on imported oil. This is tested to a further extent in the following analysis. The grid-search procedure returns a threshold $\gamma = 0.22$ that is virtually identical to the baseline estimate, and the bootstrap confidence bands for the two-regime IRFs overlap closely with the baseline bands.

6.1.2 Energy-Dependence Interaction

A central concern is that the institutional threshold may partially proxy for energy import dependence rather than institutional capacity per se. Countries with weaker institutions may be more exposed to supply-side oil shocks, and if high-institution countries systematically import less energy, the observed asymmetry in fiscal responses could reflect differential exposure to the shock rather than differences in state capacity.

Table 3: Import interaction coefficients (Ψ) — selected horizons

Horizon	Low-institution regime			High-institution regime		
	Ψ_{low}	95% CI	p	Ψ_{high}	95% CI	p
1	0.0017	[-0.0019, 0.0046]	0.240	0.0024	[-0.0018, 0.0077]	0.276
2	0.0010	[-0.0035, 0.0056]	0.486	0.0030	[-0.0019, 0.0093]	0.242
4	0.0041	[-0.0009, 0.0109]	0.096	0.0008	[-0.0063, 0.0097]	0.832
6	0.0026	[-0.0038, 0.0063]	0.374	-0.0013	[-0.0095, 0.0079]	0.770
8	0.0040	[-0.0007, 0.0093]	0.090	-0.0021	[-0.0083, 0.0036]	0.410
10	0.0011	[-0.0034, 0.0055]	0.446	0.0021	[-0.0025, 0.0069]	0.290

Notes: Ψ_{low} and Ψ_{high} are the regime-specific import interaction coefficients from equation (2). p -values are bootstrap-based (share of draws with opposite sign to point estimate). 95% CIs obtained from a country-block bootstrap; an interval spanning zero indicates Ψ is indistinguishable from zero. Magnitude = $\Psi \times \text{SD}(\text{energy imports})$, where $\text{SD} = 40.78$ pp of GDP.

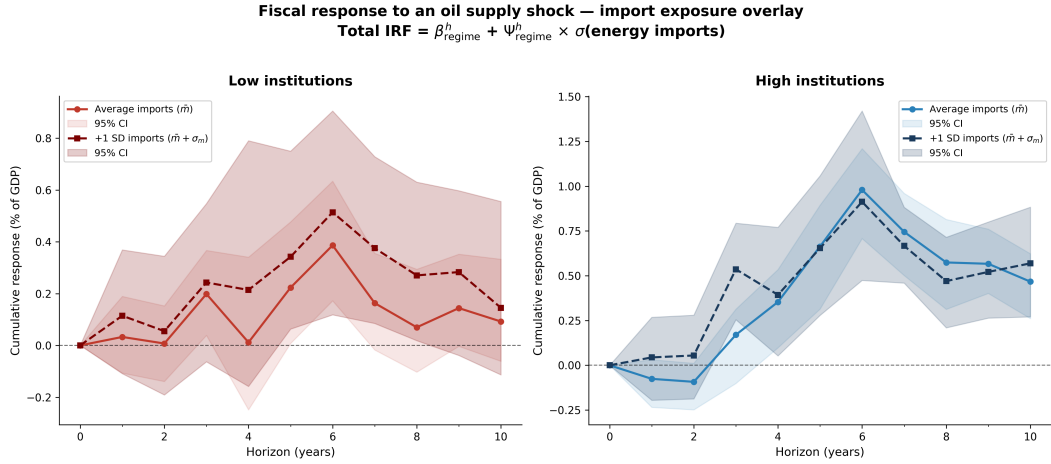


Figure 6: Import Exposure Interacted Local Projection

The energy-interacted impulse responses confirm that the institutional threshold does not proxy for oil dependence. Within each regime, the response at average imports and the response at one standard deviation above average are nearly identical, and the difference falls well within the bootstrap confidence bands. The Ψ coefficients are statistically indistinguishable from zero in both regimes, indicating that import dependence does not meaningfully alter the fiscal response once the institutional regime is controlled for. The institutional IRFs β_{low} and β_{high} are unchanged relative to the baseline, confirming that the asymmetric fiscal adjustment identified in the baseline model reflects differences in institutional capacity rather than differential exposure to oil supply shocks.

6.1.3 Time Trend

The IRFs for both the low and high-institution regimes pictured in the appendix remain virtually identical to the extended control estimates, tracking the original point estimates closely. This insensitivity to the inclusion of a linear time trend indicates that the institutional asymmetry is not an artefact of a common linear trend in the data, and it eliminates the concern that the well-documented decline in the macroeconomic impact of oil supply shocks (Baumeister and Peersman 2013) could be driving the results through a spurious correlation with institutional quality.

6.1.4 Real Variables

This final robustness check ensures that the estimated institutional asymmetry reflects deliberate fiscal behaviour rather than nominal re-valuation. The IRFs (pictured in the appendix) remain consistent in direction, magnitude, and statistical significance with the baseline estimates.

This stability confirms that the institutional gap is driven by real fiscal adjustments, rather than issues with the specification of the baseline model or any nominal constructs in the fiscal ratio. This robustness test is conducted for all other specifications in the paper (revenue, expenditure, and debt variables), yielding similar findings that are all reported in the appendix.

7 Smooth Transition Threshold

The baseline binary threshold established that fiscal responses differ markedly between low- and high-institution countries. That specification, however, forces all countries into two discrete regimes and discards any within-regime variation. The smooth-transition local projection relaxes this constraint by allowing the impulse response to vary continuously with the annual institutional index z_{it} .

Table 4: Smooth-Transition Local Projection: Estimated Parameters and Implied Responses

	Baseline controls	Extended controls
<i>Panel A: Transition function parameters</i>		
Optimal center c^*	-0.943	-0.943
95% CI for c^*	[-1.853, 1.333]	[-1.853, 1.333]
Transition speed λ^*	59.255	59.255
95% CI for λ^*	[0.594, 100.000]	[2.565, 100.000]
Transition width	0.037	0.037
Transition type	Sharp ($\lambda > 3$)	Sharp ($\lambda > 3$)
<i>Panel B: Logistic weight F at percentiles of lagged institutional quality</i>		
F at p_{10}	0.000	0.000
F at p_{25}	0.035	0.035
F at p_{50}	1.000	1.000
F at p_{75}	1.000	1.000
F at p_{90}	1.000	1.000
<i>Panel C: Implied cumulative primary balance responses at $h = 4$ (% GDP)</i>		
Low-institution mean	-0.185	-0.181
High-institution mean	0.157	0.148
Difference (high - low)	0.341	0.329

Notes: c^* denotes the optimal transition center and λ^* the transition speed, both estimated via grid search minimizing the sum of squared residuals across horizons $h = 0, \dots, 4$. Panel C reports the implied cumulative primary balance response (percentage points of GDP) evaluated at the mean institutional quality of countries below and above c^* , respectively. Bootstrap 95% confidence intervals are obtained from 1,000 country-block resamples.

The grid-search estimates a center $c^* = -0.943$ and a transition speed $\lambda^* = 59.3$. The center lies near the 25th percentile of the institutional distribution, well below the sample median. The large λ implies a near-discrete transition, as the distance between the 25th and 75th percentiles of the logistic function is only 0.037 units of the standardized index. The F-values confirm this sharpness, as the 25th percentile a country receives a weight of only 3.5% on the high-regime coefficient, while at the median the weight is already 100%. In effect, the STLP collapses almost exactly to the binary threshold model, with the break occurring at a relatively low level of institutional quality.

This near-discrete transition and the low location of the center are consistent with the composition of the sample. A large share of the observations belong to advanced

economies, including Western Europe, the United States, and developed East Asian countries, which score very highly on the composite institutional index. The relatively sparse middle of the institutional distribution means that even a smooth-transition function reduces to a nearly discrete step, separating the institutionally weak countries from a broad, homogeneous block of moderate-to-high-institution economies.

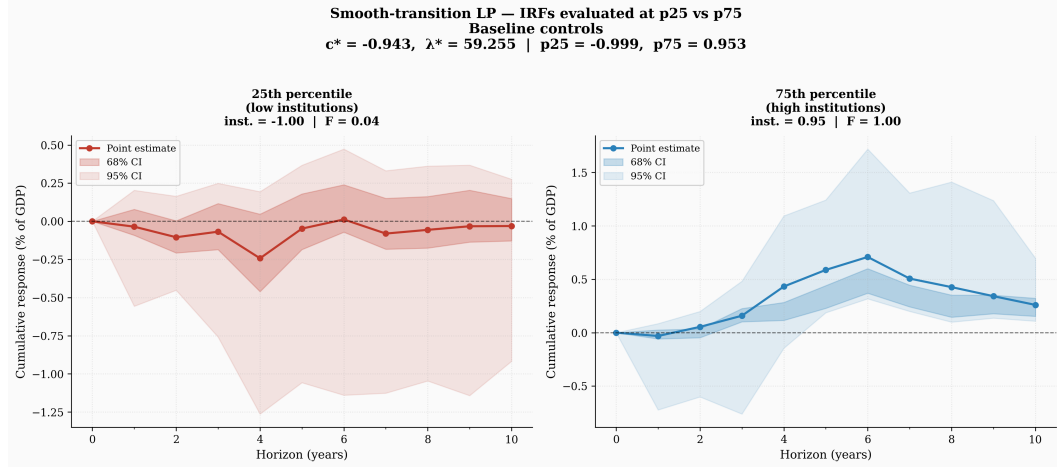


Figure 7: Baseline Smooth Transition Local Projections Quartile IRFs

The impulse responses for the regimes trace the broad pattern of the binary-threshold IRFs, as the low-extreme response hovers around zero, while the high-extreme response turns positive and remains significant out to the ten-year horizon. This convergence confirms that the binary split, despite its simplicity, already captured the broad average pattern. Thus, the STLP, and its robustness checks included in the appendix, does not overturn the earlier finding but reinforces it, as the data strongly prefers a two-regime world with a sharp divide.

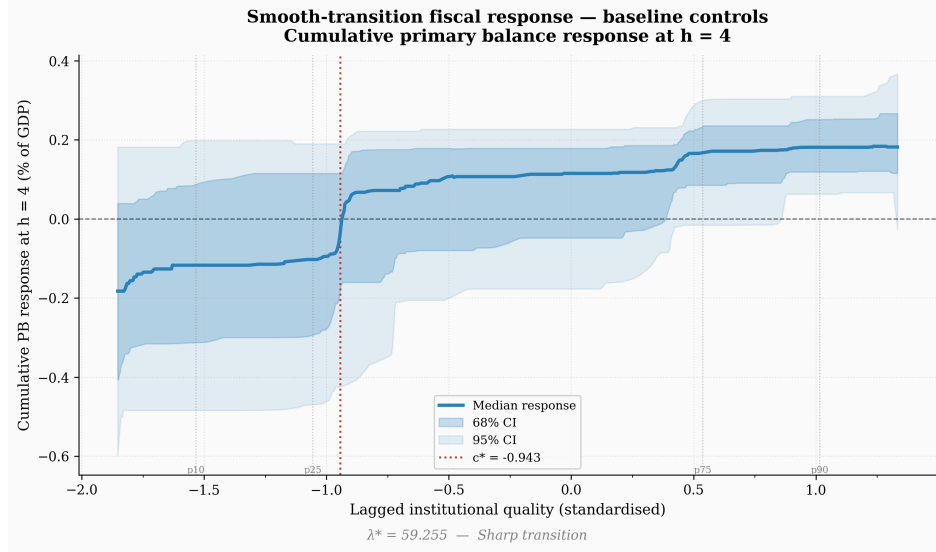


Figure 8: STLP Non-Parametric Lowess Curve

Another interesting feature of the non-parametric lowess curve is a second upward inflection at $z \approx 0.5$, just above the top quartile. While the main institutional divide is already crossed at a much lower level, this secondary rise suggests that the most advanced institutional environments, characterised by exceptionally strong rule of law, very low corruption, and highly professional bureaucracies, generate an extra primary-balance surplus beyond what the main threshold already delivers. The pattern reinforces the overall gradient while suggesting that there are more non-linearities in the institutional spectrum that a simple two-regime model cannot fully capture.

8 Fiscal Decomposition

The baseline primary-balance results demonstrated that high-institution countries systematically improve their fiscal position after an exogenous oil supply shock, whereas low-institution countries show no significant response at the 95% confidence level. The primary balance is only a summary measure, and cannot reveal whether the improvement is achieved by cutting spending, raising taxes, borrowing, or some combination. The fiscal decomposition aims to identify the main transmission channels of fiscal policy by examining the responses of expenditure, revenue (both tax and general), and public debt.

Political-Economy Hypotheses

1. **Voracity Effect:** In low-institution countries, fragmented interest groups should resist spending cuts, while expenditure should therefore be sticky. In high-institution countries, governments can impose fiscal restraint.

2. State Capacity: High-institution countries have the enforcement capacity to raise offsetting revenue, while low-institution countries, with weak tax administrations and large informal sectors, should see revenues fall.
3. Fiscal Space: High-institution countries can borrow to smooth the shock, while low-institution countries face binding borrowing constraints and may be forced to deleverage.

8.1 Government Expenditure

The expenditure IRFs show a decline in both institutional regimes. In low-institution countries, the cumulative fall reaches -0.27 percentage points after six years, and remains significantly negative through to horizon ten. In high-institution countries, expenditure also contracts, with a cumulative decline of -0.3 percentage points at horizon six, and similar magnitudes thereafter. Thus, the impulse responses do not support the voracity hypothesis set forth by Tornell and Lane (1999). Expenditure falls significantly in both institutional regimes, and the cumulative declines are statistically indistinguishable. This means there is little to no heterogeneity in the expenditure responses of the two institutional regimes, at least the ones generated by the baseline γ threshold.

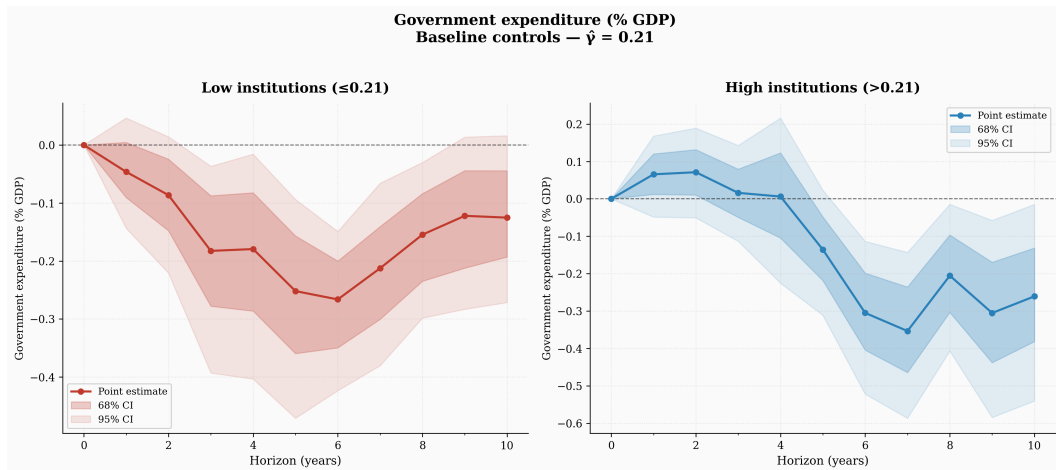


Figure 9: Government Expenditure IRFs

This is likely due to the nature of the oil supply shock for importers. Adverse oil supply disruptions raise the cost of key imported inputs, depressing real incomes, and slowing economic activity. For most governments, this reduces the nominal base from which expenditure is scaled, and puts downward pressure on spending categories. This means fuel subsidies may become too expensive to maintain, even for institutionally-

robust nations, so public investment is postponed. This aggregate negative income effect overwhelms the institutional forces that would protect expenditure.

It must be mentioned, however, that the voracity effect was conceived in the context of positive windfall, and may be less relevant when the shock is adverse, as there is no surplus to fight over only a shared need for government retrenchment.

This does not mean that institutions play no role, as the composition of expenditure cuts could still differ systematically across regimes. The aggregate data employed by this thesis is unable to provide the granular detail needed to test these channels, but that is a natural extension that could determine the margins through which expenditure is cut vis-à-vis oil supply shocks. It is also possible that high-institution countries achieve a given nominal cut with less social disruption, because stronger safety nets and more efficient public administration allow a smoother adjustment. For now, the evidence suggests that the immediate spending response to an adverse oil supply shock is driven by the common shock, not by institutional quality.

8.2 Tax Revenue and General Government Revenue

The model results show that tax revenue in high-institution countries increases modestly for several years, and fades in the medium term. This suggests that governments may adopt temporary measures such as surcharges or temporary levies that are politically easier to implement but are eventually unwound.

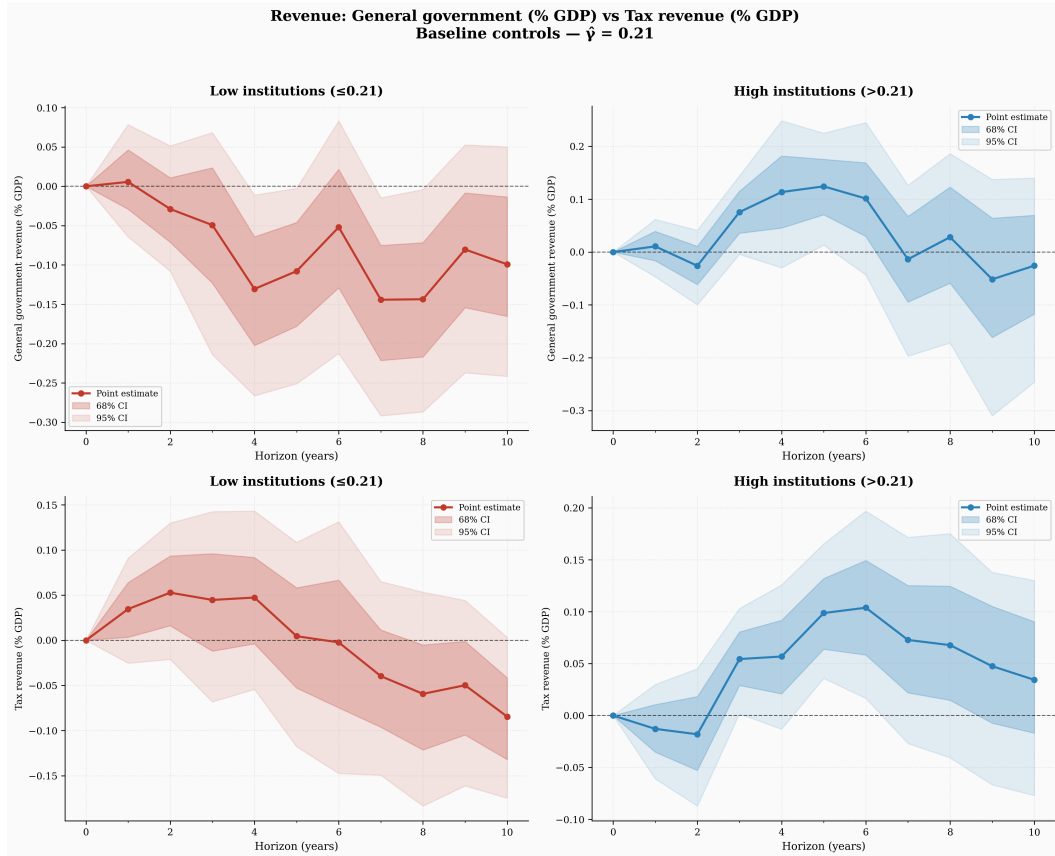


Figure 10: General Government and Tax Revenues IRFs

In low-institution economies, the progressive shrinking of the effective tax base can be explained by the mechanism identified by Kleven et al. (2016). These authors show that taxable income elasticities are high where enforcement capacity is low. Three forces drive this erosion. First, as the shock deepens and economic activity contracts, formal-sector firms shift into the informal economy to cut costs, moving transactions beyond the reach of the tax authorities. Second, the administrative machinery of low-capacity states, already stretched, cannot sustain the extra enforcement effort required to keep compliance from slipping. Third, as political pressure to provide relief through measures such as fuel subsidies, tax exemptions, or temporary tax holidays intensifies, governments face increasing demands from influential interest groups. In institutionally weaker settings, authorities are generally less able to withstand such pressures. Together, these factors widen the gap between the statutory and effective tax bases, reducing revenue collection.

This evidence broadly confirms the state-capacity hypothesis, but with a timing nuance: the capacity does not bind instantly, but rather tightens as the shock persists. This delayed response is also consistent with the longer time horizons required for the implementation and transmission of fiscal policy. This provides more detailed informa-

tion for the hypothesis, as it suggests that capacity constraints are not instantaneous but gradual. The initial spike likely reflects transient behavioral and political responses, while the medium and long term evolution in each regime reveal the structural inability of low-institution economies to sustain revenue mobilization. This points to the idea that weak institutions produce a short-term impression of resilience that gives way to underlying fiscal fragility.

The general government revenue provides a broader lens because it includes non-tax revenues that may react to oil-supply shocks. Here the asymmetry is even more pronounced, as high-institution countries show no significant decline in total government revenue, while low-institution countries suffer a persistent and significant cumulative fall. This divergence implies that the revenue base in weak-institution settings is more vulnerable to the indirect effects of an oil shock, as state-owned energy companies may be forced to absorb losses, trade-related revenues may shrink, and the informality that follows economic contraction may hinder non-tax revenues. In contrast, high-institution economies appear to have the administrative and fiscal capacity to maintain overall revenue stable even when some components come under pressure.

Taken together, the revenue evidence supports the state-capacity hypothesis, but it identifies a different channel than the predictions laid out by theory. Total revenues, not just tax revenues, distinguish institutional regimes. High-institution countries do not specifically engineer a large tax windfall but rather prevent a revenue collapse through a superior state-capacity, while low-institution countries see their fiscal base erode on multiple fronts.

8.3 Debt

High-institution countries respond to the adverse oil shock by increasing public debt significantly and persistently. This confirms the idea that financial markets treat these countries' debt as safe, allowing them to issue bonds at favourable rates even in the aftermath of a supply-side oil shock. The debt increase can be interpreted as a fiscal smoothing response, with governments financing temporary revenue shortfalls and expenditure needs through borrowing. Low-institution economies, on the other hand, see no initial change in debt in the first periods, followed by a significant decline.

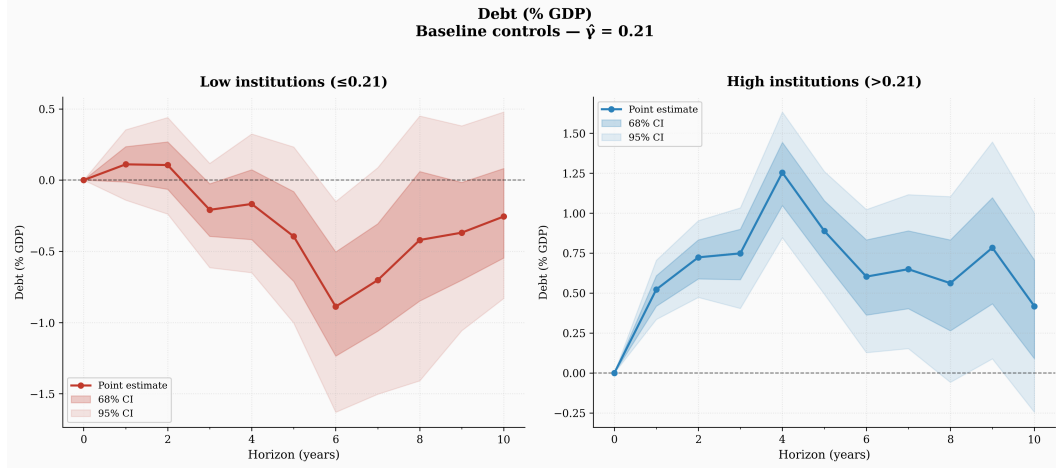


Figure 11: Government Debt IRFs

This asymmetry provides strong support for the predictions of the political-economy literature. Two related but distinct concepts help explain the pattern. The first, debt intolerance explains why low-institution countries could lose access to capital markets, long before debt-sustainability thresholds are breached (Reinhart et al. 2003). The second, fiscal space (Aizenman et al. 2019) broadens the perspective by capturing the overall capacity of a government to run expansionary fiscal policy without endangering debt sustainability, a capacity that is strongly conditioned by institutional quality and sovereign credit ratings. Thus, high-fiscal-space countries are able to borrow to cushion external shocks, while low-fiscal space countries cannot. Both predictions are present in this study, as an adverse oil supply shock triggered increased debt in institutionally robust countries, which use the borrowing capacity to smooth the shock. Meanwhile, the stagnant and declining debt ratio in institutionally weak economies is caused by their binding borrowing constraints. The falling debt ratio in the low regime can therefore be seen as an indicator of credit constraint, rather than fiscal discipline or prudent policy.

This asymmetry may go beyond simple cumulative dynamics, and may impact the conditions and rates of the debt incurred. Economies with poor institutions may borrow in foreign currency, on shorter maturities, or at punitive rates, making their debt dynamics vastly different from those of other economies. This potential debt-composition asymmetry also warrants further study, as the aggregate debt-to-GDP ratio masks the qualitative and granular differences.

The real-variable decomposition in the appendix confirms that the asymmetries found in the baseline specification are genuine and not caused by fluctuations in nominal variables. As pictured in the appendix, real expenditure falls comparably in both regimes, real debt rises strongly only for high-institution countries, and real tax revenue

shows a temporary gain for high-institution countries that is not matched by the low-institution counterpart. Institutional asymmetries in revenue and debt therefore reflect real fiscal capacity, not nominal scaling.

8.4 Financial Transmission Mechanism

Although not every hypothesis advanced in the political-economy literature was supported by the data, the analysis yields a clear and valuable insight: institutional quality shapes fiscal responses to an adverse oil supply shock primarily through the financing side. High-institution countries maintain a stable revenue base, mobilize additional tax receipts, and borrow substantially to absorb the shock. This capacity is rooted in the credibility that financial markets attach to their institutional framework. Low-institution countries, by contrast, suffer a shrinking revenue base, face binding borrowing constraints, and are forced to deleverage, with limited fiscal space and restricted market access.

These findings refine the predictions of political-economy theory in two important respects. First, the voracity effect does not appear to operate on the expenditure side in the context of a negative supply-side shock for oil importers. Second, the institutional divide is most visible not in how much governments spend, but in how they finance the adjustment. The capacity to raise revenue and to borrow on favourable terms is what separates the fiscal trajectories of institutionally strong and weak countries. This finding suggests that future research on fiscal responses to external shocks should pay closer attention to the liability side of the public balance sheet and to the political and institutional foundations of market access.

9 Institutional Decomposition

The table below reports the point estimates of the moderation effect for the ten institution-fiscal pairs with the largest absolute differences in the cumulative response at $h = 4$. The moderation effect is defined as the difference between the impulse response at the 90th and 10th percentiles of the respective institutional sub-index.

Table 5: Institutional moderation effects on fiscal outcomes (top 10 pairs)

Institution	Fiscal variable	Δ (h=4)	Low-inst.	High-inst.	λ
Control of corruption	Debt	1.448	-0.300	1.148	19.3
Democracy	Debt	1.333	0.099	1.433	8.5
Rule of law	Debt	1.295	-0.150	1.145	100.0
Property rights	Debt	1.271	0.150	1.421	100.0
Bureaucratic quality	Debt	1.185	0.205	1.389	100.0
Transparent laws	Debt	1.151	-0.047	1.104	43.9
Property rights	Primary bal.	0.446	-0.291	0.155	43.9
Democracy	Primary bal.	0.436	-0.268	0.168	19.3
Rule of law	Primary bal.	0.381	-0.214	0.167	100.0
Property rights	Expenditure	0.338	-0.207	0.131	43.9

Note: Δ is the difference in cumulative response (percentage points of GDP) between the 90th and 10th percentile of the institutional sub-index. Low-inst. and High-inst. are the respective cumulative responses at horizon 4. λ is the smoothness parameter of the logistic transition function; a value of 100 indicates a near-discrete jump. All estimates are point estimates from a smooth-transition local projection with two lags of GDP growth and CPI inflation. Bootstrap p-values < 0.05 for all pairs except possibly the bottom three (borderline).

9.1 Dominant Adjustment Channel

the difference in the debt-to-GDP response between a country at the 90th and one at the 10th percentile of a given sub-index ranges from 1.15 percentage points (for transparent laws) to 1.45 percentage points (for control of corruption). In the high-institution scenario, the debt ratio rises by ~ 1.4 percentage points above its pre-shock level, whereas in the low-institution scenario the effect is economically negligible. This asymmetry indicates that institutional quality governs a government's ability to borrow in the wake of an adverse energy shock, and it does so across all the institutional dimensions considered. The three sub-indices with the largest point estimates are control of corruption, democracy and rule of law. These three dimensions coincide with the indicators that international investors and rating agencies emphasize when assessing sovereign creditworthiness, consistent with the view that property rights, rule of law, and control of corruption are the institutional pillars of market access and fiscal space (Alesina, Campante, et al. 2008; Arezki et al. 2017).

Notably, the smoothness parameter λ reaches the upper bound of the search grid ($\lambda = 100$) for rule of law, property rights, and bureaucratic quality when debt is the outcome, implying that the transition is a drastic, almost binary jump. This confirms

that the binary-threshold specification used in the baseline is well suited to capture the key institutional variation that drives debt dynamics.

9.2 Foremost Individual Moderators

Among the six sub-indices, control of corruption displays the single largest moderation effect, followed by democracy and rule of law. The differences between these top three indicators are not statistically significant for pairwise comparisons, suggesting that they act as complementary pillars of fiscal credibility. Notably, all three dimensions produce a positive high-institution response and a near-zero or slightly negative low-institution response, reinforcing the conclusion that debt management depends critically on a state’s credibility and accountability in financial markets.

It is also worth noting that tax revenue does not appear among the ten largest moderation effects. This does not imply that institutions are irrelevant for tax collection, but rather suggests that the tax-revenue response is driven by the joint effect of several institutional dimensions, as captured by the composite index, rather than a single institutional variable.

9.3 Thresholds in Property rights, Rule of Law, and Bureaucratic Quality

The estimated smoothness parameter λ reaches the upper bound of the grid for property rights, rule of law, and bureaucratic quality when the outcome is debt. This implies that the logistic function is a step function, meaning that the fiscal response changes abruptly once a critical level of the indicator is crossed. The finding validates the binary-threshold specification used in the baseline, but it also provides more detailed information about the source of the debt-dynamic asymmetry found in the fiscal decomposition.

This pattern has a natural economic interpretation: for property rights, rule of law, and bureaucratic quality, a country’s ability to incur debt does not appear to be a smoothly varying attribute, but rather an almost binary classification. One candidate explanation is that international capital markets assess a government’s creditworthiness in discrete manners, such as through credit ratings, formal investment-grade thresholds, and informal reputational benchmarks. Institutional features such as the enforceability of property rights and the predictability of the rule of law are among the most visible signals that foreign investors and rating agencies use to classify sovereigns (Cantor and Packer 1996). Once these indicators fall below a critical level, the market effectively labels a country as “unreliable” and withholds credit, producing the flat debt response

observed for the low-institution regime. Above that threshold, the country has a reputation as a safe borrower, allowing the state to incur debt to offset the effects of an adverse oil supply shock.

These findings provide further evidence in favour of the “debt intolerance” hypothesis (Reinhart et al. 2003). The binary transition reflects the dynamics of countries with a history of default and weak institutions that lose market access long before debt-sustainability limits are reached. A country whose property rights and rule of law fall below a critical floor is treated by creditors as inherently risky, irrespective of its current debt level. Conversely, countries above this binary threshold enjoy the benefits of “fiscal space” that strong institutional frameworks provide.

The fact that the step-function behaviour is found for property rights, rule of law, and bureaucratic quality and not for control of corruption or democracy is also consistent with how credit markets operate. Rating agencies and international bondholders place particular weight on the legal and administrative infrastructure that guarantees contracts will be honoured and that the government itself is bound by predictable rules (North 1990; Besley and Persson 2011). A government that can credibly commit to respecting property rights and maintaining a competent, rule-bound bureaucracy is one that can credibly commit to repaying its debts. The binary shape of the transition suggests the presence of a “reputational precipice”, where once institutional quality drops below the critical value, the loss of market access is abrupt rather than gradual.

10 Policy Implications

The results carry practical policy implications for both national governments and international financial institutions. Building credible property rights, predictable legal systems, and competent public administrations is not merely a long-run development goal, but also a robust tool for short-run macroeconomic resilience. When energy shocks occur, countries that have robust institutions have access to a wider range of fiscal tools to mitigate adverse macroeconomic effects. The reputational aspect of institutions underpins the ability to respond to oil shocks and to leverage fiscal policy. Thus, institutional quality should not be treated as a second-order feature of a country’s risk profile but as a primary determinant of its capacity to absorb shocks and stabilise public finances. For governments, the implication is that reforms aimed at strengthening property rights, bureaucratic quality, and the rule of law are also investments in strengthening fiscal capacity.

11 Caveats and Extensions

Several limitations of this thesis merit further investigation. First, the analysis relies on aggregate fiscal data, which cannot distinguish between different types of expenditure cuts or tax increases. Furthermore, a great deal of high-frequency volatility is lost by working at an annual frequency, leaving out potentially informative quarterly and even monthly dynamics. Studying micro-level data on budget composition, or firm-level data on tax compliance, could provide information on these margins.

Second, the debt-to-GDP ratio is a summary measure that masks heterogeneity in the composition of sovereign liabilities. Low-institution countries may borrow in foreign currency, at shorter maturities, or at varying interest rates, making their true fiscal position more or less vulnerable than the aggregate ratio suggests. Incorporating data on debt structure and credit-default-swap spreads could capture these qualitative differences and would be highly informative.

Lastly, the oil-supply shock is a single exogenous disturbance. It would be informative to test whether the same institutional mechanisms condition the fiscal response to other external shocks, such as gas supply-side shocks or other commodity-price movements, to determine whether the patterns are inherent to oil as a commodity or reflect a broader phenomenon. The aggregation of the shock from monthly to yearly is also a potential drawback that dilutes much of the informative monthly variation in the original series, although this is inevitable given the yearly frequency of most national accounts.

12 Conclusion

Institutions are often thought of as the slow-moving foundations of long-run prosperity. This thesis has shown that they are also key determinants of the fiscal tools available to respond to oil shocks. The aim of the analysis was to determine whether institutional quality conditions the fiscal response to exogenous oil-supply shocks. The evidence examined here has shown that it does. An economy's state capacity, accountability, and ability to enforce the rule of law determine its capacity to leverage fiscal policy when global oil markets are disrupted. While some channels, notably government expenditure, do not display strong asymmetric dynamics across institutional regimes, a state's ability to raise revenues and to borrow in order to offset the adverse effects of crises is significantly conditioned by its institutional quality.

The analysis further revealed that the specific institutions governing a country's ability to enforce contracts and laws, namely property rights, the rule of law, and bureaucratic quality, are the dimensions that most decisively shape its access to debt markets. These institutions proxy a nation's credibility in global capital markets, and the data show that they operate not as a smooth gradient but as a reputational switch below which credit is withheld, and the government is forced into retrenchment. This binary logic supports the predictions of the political-economy literature, which state that countries with weak institutions, weak accountability, and weak reputation face binding debt constraints, limiting their ability to leverage fiscal policy when faced with staggering expenditure demands. Institutions therefore condition an economy's fiscal response by expanding or contracting the set of fiscal instruments it can use, and are thus a key component for policymakers to consider when they aim to improve a country's macroeconomic health and fiscal breadth. Strengthening those institutional foundations is therefore not only a matter of socioeconomic development but also of macroeconomic resilience in an energy-volatile world.

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14 Appendix

14.1 Binary Local Projection: Robustness Controls

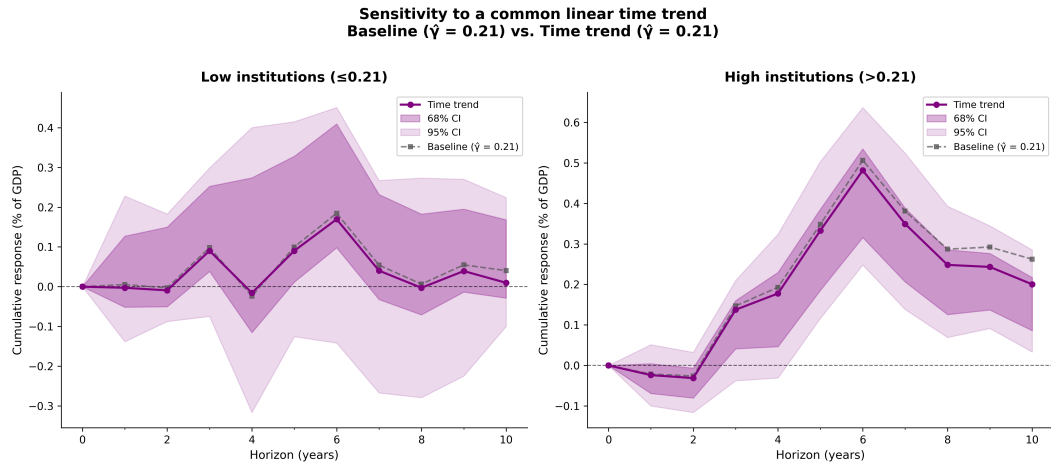


Figure 12: Time-Trend Robustness Check: Local Projection

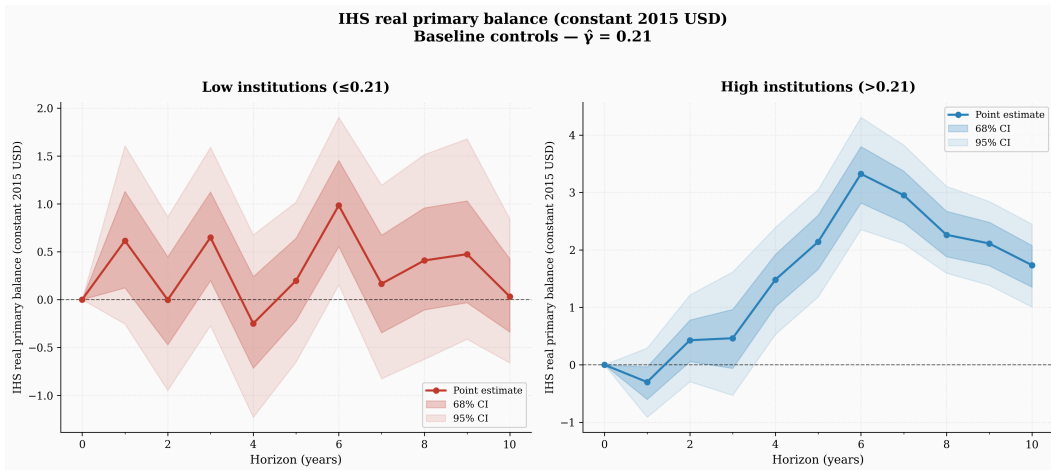


Figure 13: Robustness Check: Real Primary Balance

14.2 Smooth Transition Local Projection

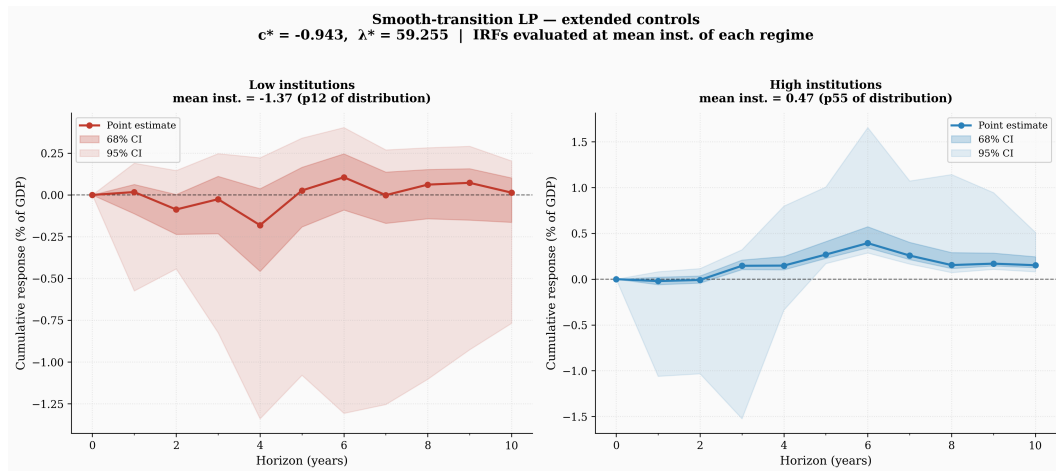


Figure 14: STLP with Extended Controls IRFs

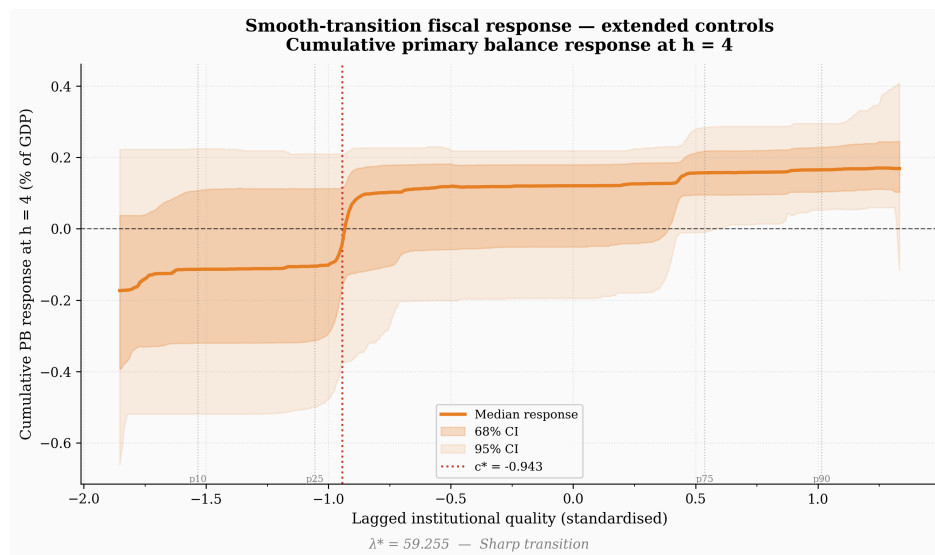
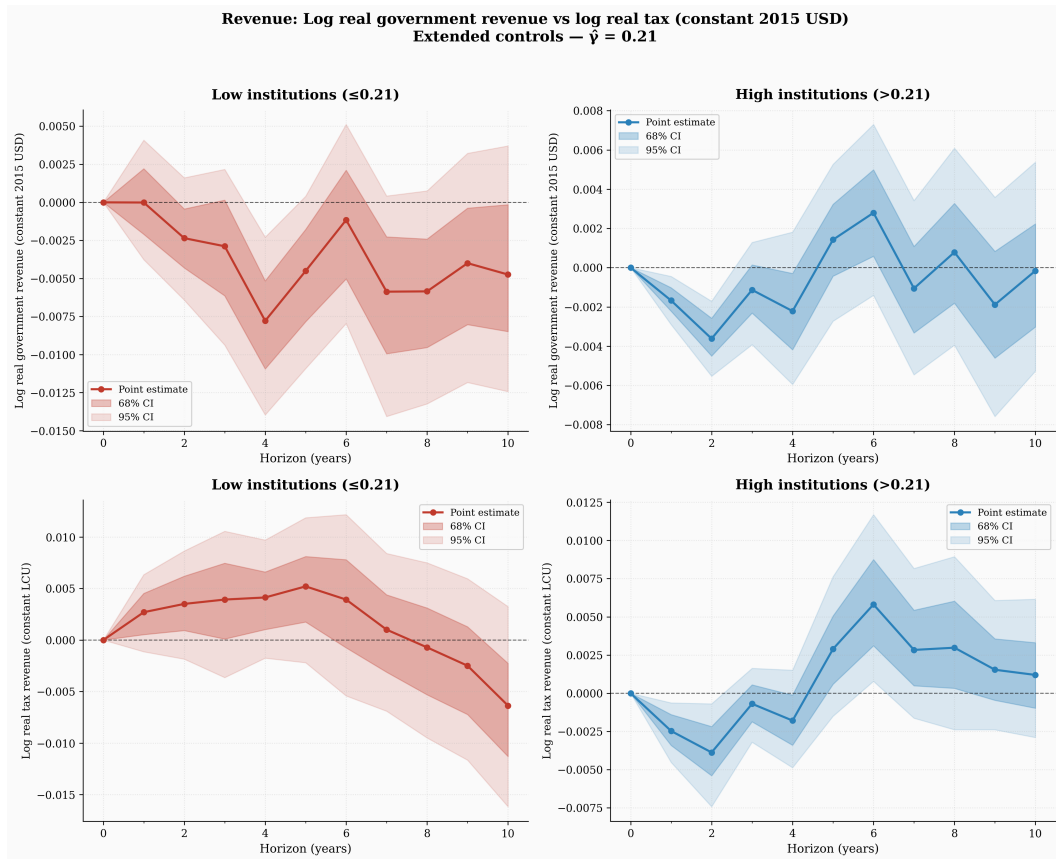
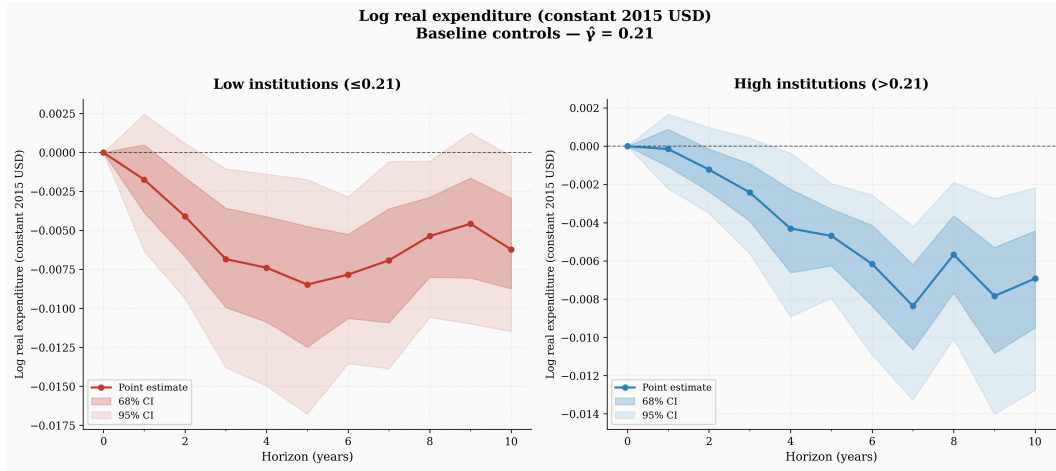


Figure 15: Non-Parametric Lowess STLP Extended Controls

14.3 Fiscal Decomposition Real-Variable Controls



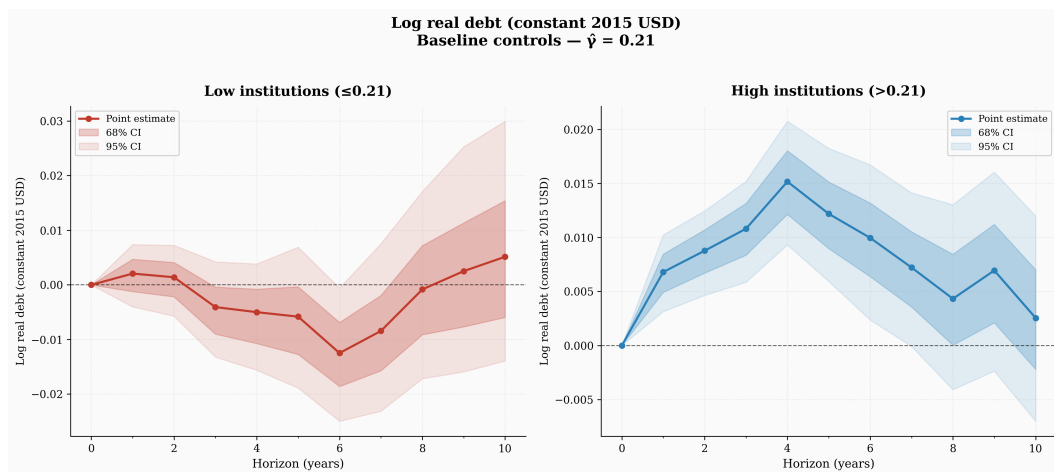


Figure 18: Real Debt IRFs

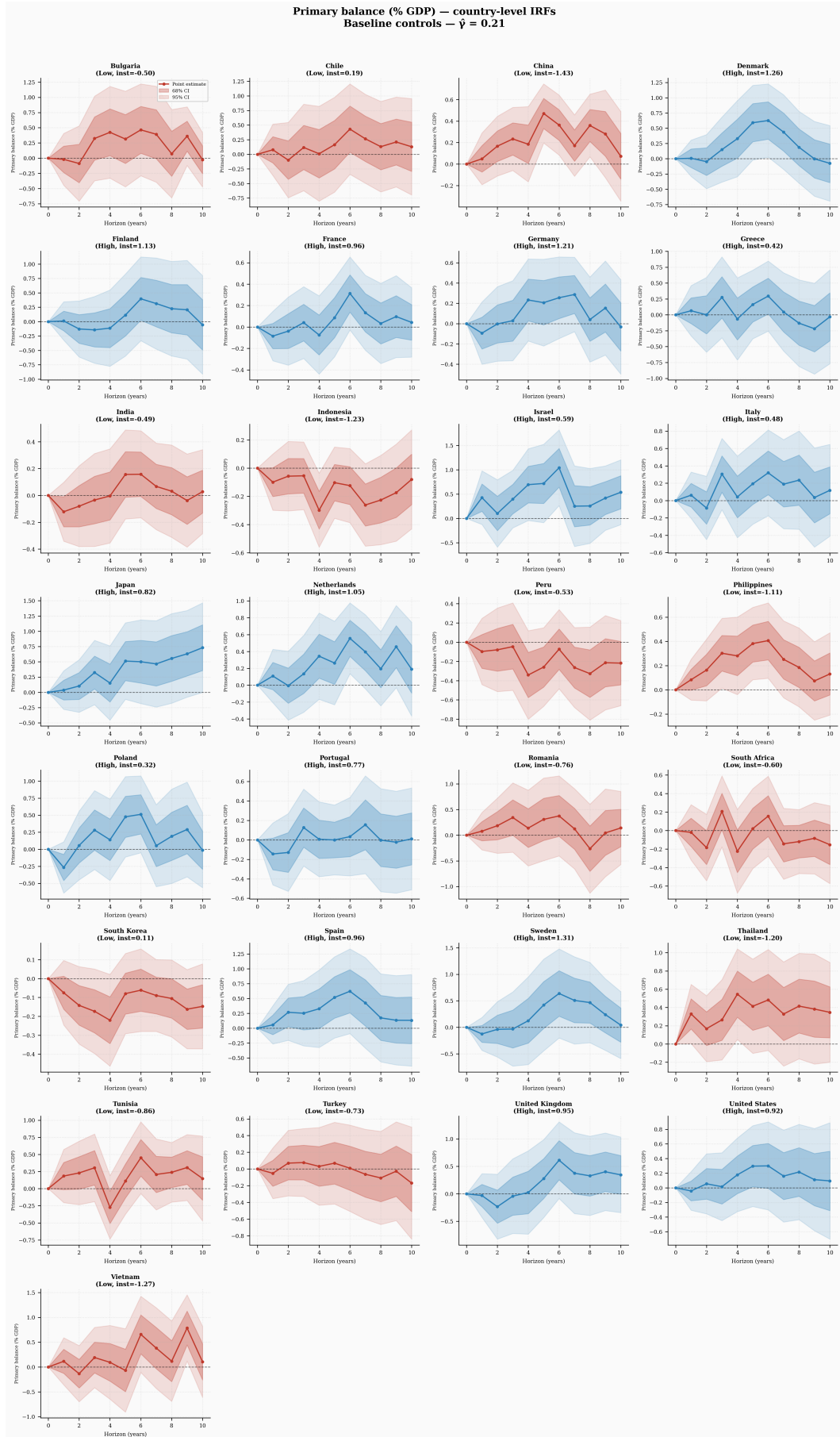


Figure 19: Country-level IRFs: Primary balance (% GDP), Baseline Controls